

fateh + 1340 R.
DIPLOMASEM:2
 (4320001/4320002)

* If $A = \begin{bmatrix} 3 & 2 & 1 \\ 0 & 1 & 0 \\ 7 & 8 & 9 \end{bmatrix}$, $B = \begin{bmatrix} -1 & -2 & 0 \\ 1 & 1 & -1 \\ 2 & 2 & 2 \end{bmatrix}$ and

$C = \begin{bmatrix} 3 & 0 & 5 \\ 6 & 9 & -1 \\ 7 & 8 & -2 \end{bmatrix}$ then find $2A - 4B + C$.

$$\Rightarrow 2A - 4B + C$$

$$= 2 \begin{bmatrix} 3 & 2 & 1 \\ 0 & 1 & 0 \\ 7 & 8 & 9 \end{bmatrix} - 4 \begin{bmatrix} -1 & -2 & 0 \\ 1 & 1 & -1 \\ 2 & 2 & 2 \end{bmatrix} + \begin{bmatrix} 3 & 0 & 5 \\ 6 & 9 & -1 \\ 7 & 8 & -2 \end{bmatrix}$$

$$= \begin{bmatrix} 6 & 4 & 2 \\ 0 & 2 & 0 \\ 14 & 16 & 18 \end{bmatrix} - \begin{bmatrix} -4 & -8 & 0 \\ 4 & 4 & -4 \\ 8 & 8 & 8 \end{bmatrix} + \begin{bmatrix} 3 & 0 & 5 \\ 6 & 9 & -1 \\ 7 & 8 & -2 \end{bmatrix}$$

$$= \begin{bmatrix} 6+4+3 & 4+8+0 & 2-0+5 \\ 0-4+6 & 2-4+9 & 0+4-1 \\ 14-8+7 & 16-8+8 & 18-8-2 \end{bmatrix}$$

$$= \begin{bmatrix} 13 & 12 & 7 \\ 2 & 7 & 3 \\ 13 & 16 & 8 \end{bmatrix}$$

* If $A = \begin{bmatrix} 1 & 2 & 1 \\ 3 & 4 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 3 & -2 & 4 \\ 1 & 5 & 0 \end{bmatrix}$ then find

'X' from $X + A + B = 0$.

$$\Rightarrow X + A + B = 0$$

$$\Rightarrow (-X) = A + B$$

$$= \begin{bmatrix} 1 & 2 & 1 \\ 3 & 4 & 2 \end{bmatrix} + \begin{bmatrix} 3 & -2 & 4 \\ 1 & 5 & 0 \end{bmatrix}$$

$$\Rightarrow (-X) = \begin{bmatrix} 4 & 0 & 5 \\ 4 & 9 & 2 \end{bmatrix}$$

$$\Rightarrow X = \begin{bmatrix} -4 & 0 & -5 \\ -4 & -9 & -2 \end{bmatrix}$$

* If $A = \begin{bmatrix} 2 & -1 & 0 \\ 3 & 2 & -4 \\ 5 & 1 & 9 \end{bmatrix}$, $B = \begin{bmatrix} 17 & -1 & 3 \\ -24 & -1 & -16 \\ -7 & 1 & 1 \end{bmatrix}$ and

$4A + 3C = B$ then find matrix 'C'.

$$\Rightarrow 4A + 3C = B$$

$$\Rightarrow 3C = B - 4A$$

$$\Rightarrow 3C = \begin{bmatrix} 17 & -1 & 3 \\ -24 & -1 & -16 \\ -7 & 1 & 1 \end{bmatrix} - 4 \begin{bmatrix} 2 & -1 & 0 \\ 3 & 2 & -4 \\ 5 & 1 & 9 \end{bmatrix}$$

$$= \begin{bmatrix} 17 & -1 & 3 \\ -24 & -1 & -16 \\ -7 & 1 & 1 \end{bmatrix} - \begin{bmatrix} 8 & -4 & 0 \\ 12 & 8 & -16 \\ 20 & 4 & 36 \end{bmatrix}$$

$$= \begin{bmatrix} 17-8 & -1+4 & 3-0 \\ -24-12 & -1-8 & -16+16 \\ -7-20 & 1-4 & 1-36 \end{bmatrix}$$

$$\Rightarrow 3C = \begin{bmatrix} 9 & 3 & 3 \\ -36 & -9 & 0 \\ -27 & -3 & -35 \end{bmatrix}$$

$$\Rightarrow C = \frac{1}{3} \begin{bmatrix} 9 & 3 & 3 \\ -36 & -9 & 0 \\ -27 & -3 & -35 \end{bmatrix}$$

* If $A = \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix}$, $B = \begin{bmatrix} -1 & -2 \\ 2 & 1 \end{bmatrix}$ then prove that

$$(A+B)^T = A^T + B^T$$

$$\Rightarrow A+B = \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix} + \begin{bmatrix} -1 & -2 \\ 2 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 3-1 & 4-2 \\ 1+2 & 2+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 3 & 3 \end{bmatrix}$$

$$\Rightarrow (A+B)^T = \begin{bmatrix} 2 & 3 \\ 2 & 3 \end{bmatrix} \quad \text{--- (1)}$$

Now, $A^T = \begin{bmatrix} 3 & 1 \\ 4 & 2 \end{bmatrix}$, $B^T = \begin{bmatrix} -1 & 2 \\ 2 & 1 \end{bmatrix}$

$$A^T + B^T = \begin{bmatrix} 3 & 1 \\ 4 & 2 \end{bmatrix} + \begin{bmatrix} -1 & 2 \\ 2 & 1 \end{bmatrix}$$

$$\Rightarrow A^T + B^T = \begin{bmatrix} 3-1 & 1+2 \\ 4-2 & 2+1 \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 2 & 3 \end{bmatrix} \quad \text{--- (2)}$$

from eq. (1) & (2) $(A+B)^T = A^T + B^T$.

* If $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 \\ 2 & 1 \\ 1 & 2 \end{bmatrix}$ then

find AB and BA .

$$\Rightarrow AB = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 \\ 2 & 1 \\ 1 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 1+4+3 & 2+2+6 \\ 4+10+6 & 8+5+12 \end{bmatrix}$$

$$AB = \begin{bmatrix} 8 & 10 \\ 20 & 25 \end{bmatrix}$$

$$\Rightarrow BA = \begin{bmatrix} 1 & 2 \\ 2 & 1 \\ 1 & 2 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \downarrow$$

$$= \begin{bmatrix} 1+8 & 2+10 & 3+12 \\ 2+4 & 4+5 & 6+6 \\ 1+8 & 2+10 & 3+12 \end{bmatrix}$$

$$= \begin{bmatrix} 9 & 12 & 15 \\ 6 & 9 & 12 \\ 9 & 12 & 15 \end{bmatrix}$$

* If $A = \begin{bmatrix} 2 & -2 \\ 3 & 1 \end{bmatrix}$, $B = \begin{bmatrix} -1 & 5 \\ 4 & -3 \end{bmatrix}$ then prove

that $(AB)^T = B^T \cdot A^T$.

$$\Rightarrow AB = \begin{bmatrix} 2 & -2 \\ 3 & 1 \end{bmatrix} \cdot \begin{bmatrix} -1 & 5 \\ 4 & -3 \end{bmatrix} \downarrow$$

$$= \begin{bmatrix} -2-8 & 10+6 \\ -3+4 & 15-3 \end{bmatrix}$$

$$\Rightarrow AB = \begin{bmatrix} -10 & 16 \\ 1 & 12 \end{bmatrix}$$

$$\Rightarrow (AB)^T = \begin{bmatrix} -10 & 1 \\ 16 & 12 \end{bmatrix} \text{ --- (1)}$$

Now, $B^T = \begin{bmatrix} -1 & 4 \\ 5 & -3 \end{bmatrix}$, $A^T = \begin{bmatrix} 2 & 3 \\ -2 & 1 \end{bmatrix}$

$$\Rightarrow B^T \cdot A^T = \begin{bmatrix} -1 & 4 \\ 5 & -3 \end{bmatrix} \cdot \begin{bmatrix} 2 & 3 \\ -2 & 1 \end{bmatrix} \downarrow$$

$$= \begin{bmatrix} -2+8 & -3+4 \\ 10+6 & 15-3 \end{bmatrix}$$

$$\Rightarrow B^T \cdot A^T = \begin{bmatrix} -10 & 1 \\ 16 & 12 \end{bmatrix} \text{ --- (2)}$$

from eqn (1) & (2) $(AB)^T = B^T \cdot A^T$.

* If $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ then prove that $A^2 - 5A - 2I = 0$.

$$\Rightarrow A^2 = A \cdot A$$

$$= \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \downarrow$$

$$= \begin{bmatrix} 1+6 & 2+8 \\ 3+12 & 6+16 \end{bmatrix} = \begin{bmatrix} 7 & 10 \\ 15 & 22 \end{bmatrix}$$

$$L.H.S = A^2 - 5A - 2I$$

$$= \begin{bmatrix} 7 & 10 \\ 15 & 22 \end{bmatrix} - 5 \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} - 2 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 7 & 10 \\ 15 & 22 \end{bmatrix} - \begin{bmatrix} 5 & 10 \\ 15 & 20 \end{bmatrix} - \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 7-5-2 & 10-10-0 \\ 15-15-0 & 22-20-2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = R.H.S.$$

* If $A = \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix}$ then prove that $A^2 - 4A + 7I = 0$.

$$\Rightarrow A^2 = A \cdot A$$

$$= \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix} \cdot \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix} \downarrow$$

$$= \begin{bmatrix} 4-3 & 6+6 \\ -2-2 & -3+4 \end{bmatrix} = \begin{bmatrix} 1 & 12 \\ -4 & 1 \end{bmatrix}$$

$$L.H.S = A^2 - 4A + 7I$$

$$= \begin{bmatrix} 1 & 12 \\ -4 & 1 \end{bmatrix} - 4 \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix} + 7 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 12 \\ -4 & 1 \end{bmatrix} - \begin{bmatrix} 8 & 12 \\ -4 & 8 \end{bmatrix} + \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix}$$

$$= \begin{bmatrix} 1-8+7 & 12-12+0 \\ -4+4+0 & 1-8+7 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = R.H.S.$$

* If $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$ then prove that $A^2 - 4A - 5I = 0$.

$$\Rightarrow A^2 = A \cdot A$$

$$= \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} \downarrow$$

$$= \begin{bmatrix} 1+4+4 & 2+2+4 & 2+4+2 \\ 2+2+4 & 4+1+4 & 4+2+2 \\ 2+4+2 & 4+2+2 & 4+4+1 \end{bmatrix} = \begin{bmatrix} 9 & 8 & 8 \\ 8 & 9 & 8 \\ 8 & 8 & 9 \end{bmatrix}$$

$$L.H.S = A^2 - 4A - 5I$$

$$= \begin{bmatrix} 9 & 8 & 8 \\ 8 & 9 & 8 \\ 8 & 8 & 9 \end{bmatrix} - 4 \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} - 5 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 9 & 8 & 8 \\ 8 & 9 & 8 \\ 8 & 8 & 9 \end{bmatrix} - \begin{bmatrix} 4 & 8 & 8 \\ 8 & 4 & 8 \\ 8 & 8 & 4 \end{bmatrix} - \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 9-4-5 & 8-8-0 & 8-8-0 \\ 8-8-0 & 9-4-5 & 8-8-0 \\ 8-8-0 & 8-8-0 & 9-4-5 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$= R.H.S.$$

* If $A = \begin{bmatrix} -1 & 3 & 5 \\ 1 & -3 & -5 \\ -1 & 3 & 5 \end{bmatrix}$ then prove that $A^2 = A$.

$$\Rightarrow A^2 = A \cdot A$$

$$= \begin{bmatrix} -1 & 3 & 5 \\ 1 & -3 & -5 \\ -1 & 3 & 5 \end{bmatrix} \cdot \begin{bmatrix} -1 & 3 & 5 \\ 1 & -3 & -5 \\ -1 & 3 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 1+3-5 & -3+9+15 & -5-15+25 \\ -1-3+5 & 3+9-15 & 5+15-25 \\ 1+3-5 & -3+9+15 & -5-15+25 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & 3 & 5 \\ 1 & -3 & -5 \\ -1 & 3 & 5 \end{bmatrix} = A.$$

* If $A = \begin{bmatrix} -4 & -3 & -3 \\ 1 & 0 & 1 \\ 4 & 4 & 3 \end{bmatrix}$ then prove that

$$\text{Adj}(A) = A.$$

$\Rightarrow$ For $\text{Adj}(A)$

$$A_{11} = \begin{vmatrix} 0 & 1 \\ 4 & 3 \end{vmatrix} = 0-4 = -4 \quad A_{12} = \begin{vmatrix} 1 & 1 \\ 4 & 3 \end{vmatrix} = 3-4 = -1$$

$$A_{13} = \begin{vmatrix} 1 & 0 \\ 4 & 4 \end{vmatrix} = 4-0 = 4 \quad A_{21} = \begin{vmatrix} 3 & -3 \\ 4 & 3 \end{vmatrix} = -9+12 = 3$$

$$A_{22} = \begin{vmatrix} -4 & -3 \\ 4 & 3 \end{vmatrix} = -12+12 = 0 \quad A_{23} = \begin{vmatrix} 4 & -3 \\ 4 & 4 \end{vmatrix} = -16+12 = -4$$

$$A_{31} = \begin{vmatrix} -3 & -3 \\ 0 & 1 \end{vmatrix} = -3+0 = -3 \quad A_{32} = \begin{vmatrix} -4 & -3 \\ 1 & 1 \end{vmatrix} = -4+3 = -1$$

$$A_{33} = \begin{vmatrix} -4 & -3 \\ 1 & 0 \end{vmatrix} = 0+3 = 3.$$

$$\Rightarrow \text{matrix} = \begin{bmatrix} -4 & -1 & 4 \\ 3 & 0 & -4 \\ -3 & -1 & 3 \end{bmatrix} \Rightarrow \text{co-factor} = \begin{bmatrix} -4 & 1 & 4 \\ -3 & 0 & 4 \\ -3 & 1 & 3 \end{bmatrix}$$

$$\Rightarrow \text{Adj}(A) = \begin{bmatrix} -4 & -3 & -3 \\ 1 & 0 & 1 \\ 4 & 4 & 3 \end{bmatrix} = A.$$

* Find the inverse of $\begin{bmatrix} 3 & -1 & 2 \\ 4 & 1 & -1 \\ 5 & 0 & 1 \end{bmatrix}$

$$\Rightarrow \text{Let } A = \begin{bmatrix} 3 & -1 & 2 \\ 4 & 1 & -1 \\ 5 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow |A| = \begin{vmatrix} 3 & -1 & 2 \\ 4 & 1 & -1 \\ 5 & 0 & 1 \end{vmatrix}$$

$$= 3 \begin{vmatrix} 1 & -1 \\ 0 & 1 \end{vmatrix} + 1 \begin{vmatrix} 4 & -1 \\ 5 & 1 \end{vmatrix} + 2 \begin{vmatrix} 4 & 1 \\ 5 & 0 \end{vmatrix}$$

$$= 3(1+0) + 1(4+5) + 2(0-5)$$

$$= 3+9-10 = 2 \neq 0$$

$\Rightarrow A^{-1}$ exists.

$\Rightarrow$ For $\text{Adj}(A)$

$$A_{11} = \begin{vmatrix} 1 & -1 \\ 0 & 1 \end{vmatrix} = 1+0 = 1$$

$$A_{12} = \begin{vmatrix} 4 & -1 \\ 5 & 1 \end{vmatrix} = 4+5 = 9$$

$$A_{13} = \begin{vmatrix} 4 & 1 \\ 5 & 0 \end{vmatrix} = 0-5 = -5$$

$$A_{21} = \begin{vmatrix} -1 & 2 \\ 0 & 1 \end{vmatrix} = -1-0 = -1$$

$$A_{22} = \begin{vmatrix} 3 & 2 \\ 5 & 1 \end{vmatrix} = 3-10 = -7$$

$$A_{23} = \begin{vmatrix} 3 & -1 \\ 5 & 0 \end{vmatrix} = 0+5 = 5$$

$$A_{31} = \begin{vmatrix} -1 & 2 \\ 1 & -1 \end{vmatrix} = 1-2 = -1$$

$$A_{32} = \begin{vmatrix} 3 & 2 \\ 4 & -1 \end{vmatrix} = -3-8 = -11$$

$$A_{33} = \begin{vmatrix} 3 & -1 \\ 4 & 1 \end{vmatrix} = 3+4 = 7.$$

$$\Rightarrow \text{matrix} = \begin{bmatrix} 1 & 9 & -5 \\ -1 & -7 & 5 \\ -1 & -11 & 7 \end{bmatrix} \Rightarrow \text{matrix of co-factor} = \begin{bmatrix} 1 & -9 & -5 \\ 1 & -7 & -5 \\ -1 & 11 & 7 \end{bmatrix}$$

$$\Rightarrow \text{Adj}(A) = \begin{bmatrix} 1 & 1 & -1 \\ -9 & -7 & 11 \\ -5 & -5 & 7 \end{bmatrix}$$

$$\Rightarrow A^{-1} = \frac{\text{Adj}(A)}{|A|} = \frac{1}{2} \begin{bmatrix} 1 & 1 & -1 \\ -9 & -7 & 11 \\ -5 & -5 & 7 \end{bmatrix}.$$

* Find the inverse of $\begin{bmatrix} -1 & 2 & -3 \\ 2 & 1 & 0 \\ 4 & -2 & 5 \end{bmatrix}$

$$\Rightarrow \text{Let } A = \begin{bmatrix} -1 & 2 & -3 \\ 2 & 1 & 0 \\ 4 & -2 & 5 \end{bmatrix}$$

$$\Rightarrow |A| = \begin{vmatrix} -1 & 2 & -3 \\ 2 & 1 & 0 \\ 4 & -2 & 5 \end{vmatrix}$$

$$= -1 \begin{vmatrix} 1 & 0 \\ -2 & 5 \end{vmatrix} - 2 \begin{vmatrix} 2 & 0 \\ 4 & 5 \end{vmatrix} - 3 \begin{vmatrix} 2 & 1 \\ 4 & -2 \end{vmatrix}$$

$$= -1(5-0) - 2(10-0) - 3(-4-4)$$

$$= -5-20+24 = -1 \neq 0.$$

$\Rightarrow A^{-1}$ exists.

$\Rightarrow$ For $\text{Adj}(A)$.

$$A_{11} = \begin{vmatrix} 1 & 0 \\ -2 & 5 \end{vmatrix} = 5+0 = 5$$

$$A_{12} = \begin{vmatrix} 2 & 0 \\ 4 & 5 \end{vmatrix} = 10-0 = 10.$$

$$A_{13} = \begin{vmatrix} 2 & 1 \\ 4 & -2 \end{vmatrix} = -4-4 = -8$$

$$A_{21} = \begin{vmatrix} 2 & -3 \\ -2 & 5 \end{vmatrix} = 10-6 = 4$$

$$A_{22} = \begin{vmatrix} -1 & -3 \\ 4 & 5 \end{vmatrix} = -5+12 = 7$$

$$A_{23} = \begin{vmatrix} -1 & 2 \\ 4 & -2 \end{vmatrix} = 2-8 = -6$$

$$A_{31} = \begin{vmatrix} 2 & -3 \\ 1 & 0 \end{vmatrix} = 0+3 = 3$$

$$A_{32} = \begin{vmatrix} -1 & -3 \\ 2 & 0 \end{vmatrix} = 0+6 = 6$$

$$A_{33} = \begin{vmatrix} -1 & 2 \\ 2 & 1 \end{vmatrix} = -1-4 = -5.$$

$$\Rightarrow \text{matrix} = \begin{bmatrix} 5 & 10 & -8 \\ 4 & 7 & -6 \\ 3 & 6 & -5 \end{bmatrix} \Rightarrow \text{matrix of co-factor} = \begin{bmatrix} 5 & -10 & -8 \\ -4 & 7 & 6 \\ 3 & -6 & -5 \end{bmatrix}.$$

$$\Rightarrow \text{Adj}(A) = \begin{bmatrix} 5 & -4 & 3 \\ -10 & 7 & -6 \\ -8 & 6 & -5 \end{bmatrix}$$

$$\Rightarrow A^{-1} = \frac{\text{Adj}(A)}{|A|} = -\frac{1}{1} \begin{bmatrix} 5 & -4 & 3 \\ -10 & 7 & -6 \\ -8 & 6 & -5 \end{bmatrix}$$

* If $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 4 \\ 1 & 3 \end{bmatrix}$ then verify

$$\text{that } (AB)^{-1} = B^{-1} \cdot A^{-1}$$

$$\Rightarrow A \cdot B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 2 & 4 \\ 1 & 3 \end{bmatrix} \downarrow$$

$$AB = \begin{bmatrix} 2+1 & 4+3 \\ 0+1 & 0+3 \end{bmatrix} = \begin{bmatrix} 3 & 7 \\ 1 & 3 \end{bmatrix}$$

$$\Rightarrow |AB| = \begin{vmatrix} 3 & 7 \\ 1 & 3 \end{vmatrix} = 9 - 7 = 2 \neq 0$$

$\Rightarrow (AB)^{-1}$ is exists.

$$\Rightarrow \text{Adj}(AB) = \begin{bmatrix} 3 & -7 \\ -1 & 3 \end{bmatrix}$$

$$\Rightarrow (AB)^{-1} = \frac{\text{Adj}(AB)}{|AB|} = \frac{1}{2} \begin{bmatrix} 3 & -7 \\ -1 & 3 \end{bmatrix} \text{--- (1)}$$

$\Rightarrow$ For B^{-1}

$$|B| = \begin{vmatrix} 2 & 4 \\ 1 & 3 \end{vmatrix} = 6 - 4 = 2 \neq 0$$

B^{-1} is exists.

$$\Rightarrow \text{Adj}(B) = \begin{bmatrix} 3 & -4 \\ -1 & 2 \end{bmatrix}$$

$$\Rightarrow B^{-1} = \frac{\text{Adj}(B)}{|B|}$$

$$= \frac{1}{2} \begin{bmatrix} 3 & -4 \\ -1 & 2 \end{bmatrix}$$

$\Rightarrow$ For A^{-1}

$$|A| = \begin{vmatrix} 1 & 1 \\ 0 & 1 \end{vmatrix}$$

$$= 1 - 0 = 1 \neq 0$$

A^{-1} is exists.

$$\Rightarrow \text{Adj}(A) = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$$

$$\Rightarrow A^{-1} = \frac{\text{Adj}(A)}{|A|}$$

$$= \frac{1}{1} \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$$

Now, $B^{-1} \cdot A^{-1}$

$$= \frac{1}{2} \begin{bmatrix} 3 & -4 \\ -1 & 2 \end{bmatrix} \cdot \frac{1}{1} \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \downarrow$$

$$= \frac{1}{2} \begin{bmatrix} 3+0 & -3-4 \\ -1+0 & 1+2 \end{bmatrix}$$

$$B^{-1} \cdot A^{-1} = \frac{1}{2} \begin{bmatrix} 3 & -7 \\ -1 & 3 \end{bmatrix} \text{--- (2)}$$

from eqⁿ (1) & (2) $(AB)^{-1} = B^{-1} \cdot A^{-1}$

* Solve the equation $5x+3y=11$ and $3x-2y=-1$ using matrix method.

$$\Rightarrow 5x+3y=11$$

$$3x-2y=-1$$

$$\Rightarrow \begin{bmatrix} 5 & 3 \\ 3 & -2 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 11 \\ -1 \end{bmatrix}$$

$$A \cdot X = B$$

$$\Rightarrow X = A^{-1} \cdot B \text{--- (1)}$$

$\Rightarrow$ For A^{-1}

$$|A| = \begin{vmatrix} 5 & 3 \\ 3 & -2 \end{vmatrix} = -10 - 9 = -19 \neq 0$$

A^{-1} is exists.

$$\Rightarrow \text{Adj}(A) = \begin{bmatrix} -2 & -3 \\ -3 & 5 \end{bmatrix}$$

$$A^{-1} = \frac{\text{Adj}(A)}{|A|} = -\frac{1}{19} \begin{bmatrix} -2 & -3 \\ -3 & 5 \end{bmatrix}$$

$$\text{from (1)} \quad X = A^{-1} \cdot B$$

$$= -\frac{1}{19} \begin{bmatrix} -2 & -3 \\ -3 & 5 \end{bmatrix} \cdot \begin{bmatrix} 11 \\ -1 \end{bmatrix}$$

$$= -\frac{1}{19} \begin{bmatrix} -22+3 \\ -33-5 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \end{bmatrix} = -\frac{1}{19} \begin{bmatrix} -19 \\ -38 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -19/-19 \\ -38/-19 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$\Rightarrow x=1, y=2$$

* Solve the equation $2x+3y=1$ and $y-4x=2$ using matrix method.

$$\Rightarrow 2x+3y=1$$

$$-4x+y=2$$

$$\Rightarrow \begin{bmatrix} 2 & 3 \\ -4 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$A \cdot X = B$$

$$\Rightarrow X = A^{-1} \cdot B \text{--- (1)}$$

$\Rightarrow$ For A^{-1}

$$|A| = \begin{vmatrix} 2 & 3 \\ -4 & 1 \end{vmatrix} = 2 + 12 = 14 \neq 0$$

A^{-1} is exists.

$$\Rightarrow \text{Adj}(A) = \begin{bmatrix} 1 & -3 \\ 4 & 2 \end{bmatrix}$$

$$A^{-1} = \frac{\text{Adj}(A)}{|A|} = \frac{1}{14} \begin{bmatrix} 1 & -3 \\ 4 & 2 \end{bmatrix}$$

$$\Rightarrow \text{from (1)}$$

$$X = A^{-1} \cdot B$$

$$= \frac{1}{14} \begin{bmatrix} 1 & -3 \\ 4 & 2 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$= \frac{1}{14} \begin{bmatrix} 1-6 \\ 4+4 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{14} \begin{bmatrix} -5 \\ 8 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -5/14 \\ 8/14 \end{bmatrix}$$

$$\Rightarrow x = -\frac{5}{14}, \quad y = \frac{8}{14}.$$

* Solve the equation $3x+2y=5$ and $2x-y=1$ using matrix method.

$$\Rightarrow \begin{matrix} 3x+2y=5 \\ 2x-y=1 \end{matrix}$$

$$\Rightarrow \begin{bmatrix} 3 & 2 \\ 2 & -1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$$

$$A \cdot X = B$$

$$\Rightarrow X = A^{-1} \cdot B \text{ — (1)}$$

$$\Rightarrow \text{For } A^{-1}$$

$$|A| = \begin{vmatrix} 3 & 2 \\ 2 & -1 \end{vmatrix} = -3-4 = -7 \neq 0$$

A^{-1} is exists.

$$\Rightarrow \text{Adj}(A) = \begin{bmatrix} -1 & -2 \\ -2 & 3 \end{bmatrix}$$

$$\Rightarrow A^{-1} = \frac{\text{Adj}(A)}{|A|} = -\frac{1}{7} \begin{bmatrix} -1 & -2 \\ -2 & 3 \end{bmatrix}$$

$$\Rightarrow \text{from (1)}$$

$$X = A^{-1} \cdot B$$

$$= -\frac{1}{7} \begin{bmatrix} -1 & -2 \\ -2 & 3 \end{bmatrix} \cdot \begin{bmatrix} 5 \\ 1 \end{bmatrix}$$

$$= -\frac{1}{7} \begin{bmatrix} -5-2 \\ -10+3 \end{bmatrix}$$

$$= -\frac{1}{7} \begin{bmatrix} -7 \\ -7 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -7/-7 \\ -7/-7 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\Rightarrow x=1, \quad y=1.$$

* Solve the equation $2x+3y=6xy$ and $x-y=xy$ using matrix method.

$$\Rightarrow 2x+3y=6xy, \quad x-y=xy$$

$$\Rightarrow \frac{2x}{xy} + \frac{3y}{xy} = \frac{6xy}{xy}, \quad \frac{x}{xy} - \frac{y}{xy} = \frac{xy}{xy}$$

$$\Rightarrow \frac{2}{y} + \frac{3}{x} = 6, \quad \frac{1}{y} - \frac{1}{x} = 1$$

$$\text{Let, } \frac{1}{y} = a, \quad \frac{1}{x} = b$$

$$\Rightarrow 2a+3b=6, \quad a-b=1$$

$$\Rightarrow \begin{matrix} 2a+3b=6 \\ a-b=1 \end{matrix}$$

$$\Rightarrow \begin{bmatrix} 2 & 3 \\ 1 & -1 \end{bmatrix} \cdot \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 6 \\ 1 \end{bmatrix}$$

$$A \cdot X = B$$

$$\Rightarrow X = A^{-1} \cdot B \text{ — (1)}$$

$$\text{For } A^{-1}$$

$$|A| = \begin{vmatrix} 2 & 3 \\ 1 & -1 \end{vmatrix} = -2-3 = -5 \neq 0$$

A^{-1} is exists.

$$\Rightarrow \text{Adj}(A) = \begin{bmatrix} -1 & -3 \\ -1 & 2 \end{bmatrix}$$

$$\Rightarrow A^{-1} = \frac{\text{Adj}(A)}{|A|} = -\frac{1}{5} \begin{bmatrix} -1 & -3 \\ -1 & 2 \end{bmatrix}$$

$$\Rightarrow \text{From (1)}$$

$$X = A^{-1} \cdot B$$

$$= -\frac{1}{5} \begin{bmatrix} -1 & -3 \\ -1 & 2 \end{bmatrix} \cdot \begin{bmatrix} 6 \\ 1 \end{bmatrix}$$

$$= -\frac{1}{5} \begin{bmatrix} -6-3 \\ -6+2 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} a \\ b \end{bmatrix} = -\frac{1}{5} \begin{bmatrix} -9 \\ -4 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 9/5 \\ 4/5 \end{bmatrix} = \begin{bmatrix} 9/5 \\ 4/5 \end{bmatrix}$$

$$\Rightarrow a = 9/5, \quad b = 4/5$$

$$\text{Now } \frac{1}{y} = a = 9/5, \quad \frac{1}{x} = b = 4/5$$

$$\Rightarrow y = 5/9, \quad x = 5/4.$$

* If $y = e^x \cdot \sin x$ then find $\frac{dy}{dx}$.

$$\begin{aligned}\Rightarrow \frac{dy}{dx} &= \frac{d}{dx} (e^x \cdot \sin x) \\ &= e^x \cdot \frac{d}{dx} (\sin x) + \sin x \cdot \frac{d}{dx} (e^x) \\ &= e^x \cdot \cos x + \sin x \cdot e^x \\ \frac{dy}{dx} &= e^x (\cos x + \sin x).\end{aligned}$$

* If $y = x^3 \cdot \log x$ then find $\frac{dy}{dx}$.

$$\begin{aligned}\Rightarrow \frac{dy}{dx} &= \frac{d}{dx} (x^3 \cdot \log x) \\ &= x^3 \cdot \frac{d}{dx} (\log x) + \log x \cdot \frac{d}{dx} (x^3) \\ &= x^3 \cdot \frac{1}{x} + \log x \cdot 3x^2 \\ \Rightarrow \frac{dy}{dx} &= x^2 + \log x \cdot 3x^2.\end{aligned}$$

* If $y = \log (\sin 2x)$ then find $\frac{dy}{dx}$.

$$\begin{aligned}\Rightarrow \frac{dy}{dx} &= \frac{d}{dx} [\log (\sin 2x)] \\ &= \frac{1}{\sin 2x} \cdot \frac{d}{dx} (\sin 2x) \\ &= \frac{1}{\sin 2x} \cdot \cos 2x \cdot \frac{d}{dx} (2x) \\ \frac{dy}{dx} &= \cot 2x \cdot 2.\end{aligned}$$

* If $y = e^{2x} \cdot \sin 3x$ then find $\frac{dy}{dx}$.

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx} (e^{2x} \cdot \sin 3x) \\ &= e^{2x} \cdot \frac{d}{dx} (\sin 3x) + \sin 3x \cdot \frac{d}{dx} (e^{2x}) \\ &= e^{2x} \cdot \cos 3x \cdot \frac{d}{dx} (3x) + \sin 3x \cdot e^{2x} \cdot \frac{d}{dx} (2x) \\ &= e^{2x} \cdot \cos 3x \cdot 3 + \sin 3x \cdot e^{2x} \cdot 2 \\ \Rightarrow \frac{dy}{dx} &= e^{2x} [3 \cos 3x + 2 \sin 3x].\end{aligned}$$

* If $y = \frac{1-x^2}{1+x^2}$ then find $\frac{dy}{dx}$.

$$\begin{aligned}\Rightarrow \frac{dy}{dx} &= \frac{d}{dx} \left(\frac{1-x^2}{1+x^2} \right) \\ &= \frac{(1+x^2) \cdot \frac{d}{dx} (1-x^2) - (1-x^2) \cdot \frac{d}{dx} (1+x^2)}{(1+x^2)^2}\end{aligned}$$

$$\begin{aligned}&= \frac{(1+x^2)(0-2x) - (1-x^2)(0+2x)}{(1+x^2)^2} \\ &= \frac{2x [(1+x^2)(-1) - (1-x^2)]}{(1+x^2)^2} \\ &= \frac{2x [-1-x^2-1+x^2]}{(1+x^2)^2}\end{aligned}$$

$$= \frac{2x(-2)}{(1+x^2)^2}$$

$$\Rightarrow \frac{dy}{dx} = \frac{-4x}{(1+x^2)^2}.$$

* If $y = \frac{1+\tan x}{1-\tan x}$ then find $\frac{dy}{dx}$.

$$\begin{aligned}\Rightarrow \frac{dy}{dx} &= \frac{d}{dx} \left(\frac{1+\tan x}{1-\tan x} \right) \\ &= \frac{(1-\tan x) \cdot \frac{d}{dx} (1+\tan x) - (1+\tan x) \cdot \frac{d}{dx} (1-\tan x)}{(1-\tan x)^2} \\ &= \frac{(1-\tan x)(\sec^2 x) - (1+\tan x)(-\sec^2 x)}{(1-\tan x)^2} \\ &= \frac{\sec^2 x [(1-\tan x) - (1+\tan x)(-1)]}{(1-\tan x)^2} \\ &= \frac{\sec^2 x [1-\tan x + 1 + \tan x]}{(1-\tan x)^2}\end{aligned}$$

$$= \frac{\sec^2 x (2)}{(1-\tan x)^2}$$

$$\Rightarrow \frac{dy}{dx} = \frac{2 \sec^2 x}{(1-\tan x)^2}.$$

* If $y = \frac{a+b \sin x}{b+a \sin x}$ then find $\frac{dy}{dx}$.

$$\begin{aligned}\Rightarrow \frac{dy}{dx} &= \frac{d}{dx} \left(\frac{a+b \sin x}{b+a \sin x} \right) \\ &= \frac{(b+a \sin x) \cdot \frac{d}{dx} (a+b \sin x) - (a+b \sin x) \cdot \frac{d}{dx} (b+a \sin x)}{(b+a \sin x)^2} \\ &= \frac{(b+a \sin x)(b \cos x) - (a+b \sin x)(a \cos x)}{(b+a \sin x)^2} \\ &= \frac{\cos x [(b+a \sin x)(b) - (a+b \sin x)(a)]}{(b+a \sin x)^2}\end{aligned}$$

$$= \frac{\cos x [b^2 + ab \sin x - a^2 - ab \sin x]}{(b + a \sin x)^2}$$

$$\Rightarrow \frac{dy}{dx} = \frac{\cos x (b^2 - a^2)}{(b + a \sin x)^2}$$

* If $y = \log(\sec x + \tan x)$ then find $\frac{dy}{dx}$.

$$\Rightarrow \frac{dy}{dx} = \frac{d}{dx} [\log(\sec x + \tan x)]$$

$$= \frac{1}{\sec x + \tan x} \cdot \frac{d}{dx} (\sec x + \tan x)$$

$$= \frac{1}{\sec x + \tan x} \cdot (\sec x \cdot \tan x + \sec^2 x)$$

$$= \frac{1}{\sec x + \tan x} \cdot \sec x (\tan x + \sec x)$$

$$\Rightarrow \frac{dy}{dx} = \sec x.$$

* If $y = x^3 \cdot \sin(\log x)$ then find $\frac{dy}{dx}$.

$$\Rightarrow \frac{dy}{dx} = \frac{d}{dx} (x^3 \cdot \sin(\log x))$$

$$= x^3 \cdot \frac{d}{dx} [\sin(\log x)] + \sin(\log x) \cdot \frac{d}{dx} (x^3)$$

$$= x^3 \cdot \cos(\log x) \cdot \frac{d}{dx} (\log x) + \sin(\log x) \cdot 3x^2$$

$$= x^3 \cdot \cos(\log x) \cdot \frac{1}{x} + \sin(\log x) \cdot 3x^2$$

$$\Rightarrow \frac{dy}{dx} = x^2 \cdot \cos(\log x) + \sin(\log x) \cdot 3x^2$$

* If $y = \frac{\sin(\log x)}{x}$ then find $\frac{dy}{dx}$.

$$\Rightarrow \frac{dy}{dx} = \frac{d}{dx} \left[\frac{\sin(\log x)}{x} \right]$$

$$= \frac{x \cdot \frac{d}{dx} [\sin(\log x)] - \sin(\log x) \cdot \frac{d}{dx} (x)}{x^2}$$

$$= \frac{x \cdot \cos(\log x) \cdot \frac{d}{dx} (\log x) - \sin(\log x) (1)}{x^2}$$

$$= \frac{x \cdot \cos(\log x) \cdot \frac{1}{x} - \sin(\log x)}{x^2}$$

$$\Rightarrow \frac{dy}{dx} = \frac{\cos(\log x) - \sin(\log x)}{x^2}$$

* If $y = \log(\tan x) + \cos x + x$ then find $\frac{dy}{dx}$.

$$\Rightarrow \frac{dy}{dx} = \frac{d}{dx} [\log(\tan x) + \cos x + x]$$

$$= \frac{d}{dx} (\log \tan x) + \frac{d}{dx} (\cos x) + \frac{d}{dx} (x)$$

$$= \frac{1}{\tan x} \cdot \frac{d}{dx} (\tan x) - \sin x + 1$$

$$= \frac{1}{\tan x} \cdot \sec^2 x - \sin x + 1$$

$$\Rightarrow \frac{dy}{dx} = \frac{\sec^2 x}{\tan x} - \sin x + 1$$

* If $y = \log\left(\frac{\sin x}{1 + \cos x}\right)$ then find $\frac{dy}{dx}$.

$$\Rightarrow \frac{dy}{dx} = \frac{d}{dx} \left[\log\left(\frac{\sin x}{1 + \cos x}\right) \right]$$

$$= \frac{1}{\frac{\sin x}{1 + \cos x}} \cdot \frac{d}{dx} \left(\frac{\sin x}{1 + \cos x} \right)$$

$$= \frac{1 + \cos x}{\sin x} \cdot \left[\frac{(1 + \cos x) \cdot \frac{d}{dx} (\sin x) - \sin x \cdot \frac{d}{dx} (1 + \cos x)}{(1 + \cos x)^2} \right]$$

$$= \frac{1}{\sin x} \cdot \left[\frac{(1 + \cos x)(\cos x) - \sin x(-\sin x)}{1 + \cos x} \right]$$

$$= \frac{1}{\sin x} \cdot \left[\frac{\cos x + \cos^2 x + \sin^2 x}{1 + \cos x} \right]$$

$$= \frac{1}{\sin x} \cdot \left[\frac{\cos x + 1}{1 + \cos x} \right]$$

$$= \frac{1}{\sin x}$$

$$\Rightarrow \frac{dy}{dx} = \operatorname{cosec} x.$$

* If $y = \log \sqrt{\frac{1 - \sin x}{1 + \sin x}}$ then find $\frac{dy}{dx}$.

$$\Rightarrow y = \log \left(\frac{1 - \sin x}{1 + \sin x} \right)^{1/2}$$

$$= \frac{1}{2} \log \left(\frac{1 - \sin x}{1 + \sin x} \right)$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{2} \frac{d}{dx} \left[\log \left(\frac{1 - \sin x}{1 + \sin x} \right) \right]$$

$$= \frac{1}{2} \left[\frac{1}{\frac{1 - \sin x}{1 + \sin x}} \cdot \frac{d}{dx} \left(\frac{1 - \sin x}{1 + \sin x} \right) \right]$$

$$= \frac{1}{2} \left[\frac{\frac{1+\sin x}{1-\sin x} \cdot ((1+\sin x) \cdot \frac{d}{dx}(1-\sin x) - (1-\sin x) \cdot \frac{d}{dx}(1+\sin x))}{(1+\sin x)^2} \right]$$

$$= \frac{1}{2} \left[\frac{1}{1-\sin x} \cdot \left(\frac{(1+\sin x)(-\cos x) - (1-\sin x)(\cos x)}{1+\sin x} \right) \right]$$

$$= \frac{1}{2} \left[\frac{1}{1-\sin x} \cdot \frac{\cos x [(1+\sin x)(-1) - (1-\sin x)]}{1+\sin x} \right]$$

$$= \frac{1}{2} \left[\frac{1}{1-\sin x} \cdot \frac{\cos x [-1-\sin x-1+\sin x]}{1+\sin x} \right]$$

$$= \frac{1}{2} \left[\frac{1}{1-\sin x} \cdot \frac{\cos x (-2)}{1+\sin x} \right]$$

$$= \frac{1}{2} \left[\frac{-2 \cos x}{1-\sin^2 x} \right]$$

$$= \frac{-\cos x}{\cos^2 x}$$

$$= -\frac{1}{\cos x}$$

$$\Rightarrow \frac{dy}{dx} = -\sec x.$$

* If $y = \log [x + \sqrt{x^2 + a^2}]$ then find $\frac{dy}{dx}$.

$$\Rightarrow \frac{dy}{dx} = \frac{d}{dx} [\log (x + \sqrt{x^2 + a^2})]$$

$$= \frac{1}{x + \sqrt{x^2 + a^2}} \cdot \frac{d}{dx} (x + \sqrt{x^2 + a^2})$$

$$= \frac{1}{x + \sqrt{x^2 + a^2}} \cdot \left[1 + \frac{1}{2\sqrt{x^2 + a^2}} \cdot \frac{d}{dx} (x^2 + a^2) \right]$$

$$= \frac{1}{x + \sqrt{x^2 + a^2}} \left[1 + \frac{1}{2\sqrt{x^2 + a^2}} \cdot 2x \right]$$

$$= \frac{1}{x + \sqrt{x^2 + a^2}} \cdot \left[1 + \frac{x}{\sqrt{x^2 + a^2}} \right]$$

$$= \frac{1}{x + \sqrt{x^2 + a^2}} \cdot \left[\frac{\sqrt{x^2 + a^2} + x}{\sqrt{x^2 + a^2}} \right]$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{\sqrt{x^2 + a^2}}.$$

* If $y = x^x$ then find $\frac{dy}{dx}$.

$$\Rightarrow y = x^x$$

Taking log on both side.

$$\Rightarrow \log y = \log x^x$$

$$\Rightarrow \log y = x \cdot \log x$$

Diff. w.r.to 'x' both side.

$$\Rightarrow \frac{d}{dx} (\log y) = \frac{d}{dx} (x \cdot \log x)$$

$$\Rightarrow \frac{1}{y} \frac{dy}{dx} = x \cdot \frac{d}{dx} (\log x) + \log x \cdot \frac{d}{dx} (x)$$

$$\Rightarrow \frac{1}{y} \frac{dy}{dx} = x \cdot \frac{1}{x} + \log x \cdot (1)$$

$$\Rightarrow \frac{1}{y} \frac{dy}{dx} = 1 + \log x$$

$$\Rightarrow \frac{dy}{dx} = y [1 + \log x]$$

$$\Rightarrow \frac{dy}{dx} = x^x [1 + \log x].$$

* If $y = x^{\sin x}$ then find $\frac{dy}{dx}$.

$$\Rightarrow y = x^{\sin x}$$

Taking log on both side.

$$\Rightarrow \log y = (\log x^{\sin x})$$

$$\Rightarrow \log y = \sin x \cdot \log x.$$

Diff. w.r.to. 'x' both side.

$$\Rightarrow \frac{d}{dx} (\log y) = \frac{d}{dx} (\sin x \cdot \log x)$$

$$\Rightarrow \frac{1}{y} \frac{dy}{dx} = \sin x \cdot \frac{d}{dx} (\log x) + \log x \cdot \frac{d}{dx} (\sin x)$$

$$\Rightarrow \frac{1}{y} \frac{dy}{dx} = \sin x \cdot \frac{1}{x} + \log x \cdot \cos x$$

$$\Rightarrow \frac{1}{y} \frac{dy}{dx} = \frac{\sin x}{x} + \log x \cdot \cos x$$

$$\Rightarrow \frac{dy}{dx} = y \left[\frac{\sin x}{x} + \log x \cdot \cos x \right]$$

$$\Rightarrow \frac{dy}{dx} = x^{\sin x} \left[\frac{\sin x}{x} + \log x \cdot \cos x \right].$$

* If $y = \sin x^{\tan x}$ then find $\frac{dy}{dx}$.

$$\Rightarrow y = \sin x^{\tan x}$$

Taking log on both side.

$$\Rightarrow \log y = \log \sin x^{\tan x}$$

$$\Rightarrow \log y = \tan x \cdot \log \sin x.$$

Diff. w.r.to 'x' both side.

$$\Rightarrow \frac{d}{dx} (\log y) = \frac{d}{dx} (\tan x \cdot \log \sin x)$$

$$\Rightarrow \frac{1}{y} \frac{dy}{dx} = \tan x \cdot \frac{d}{dx} (\log \sin x) + \log \sin x \cdot \frac{d}{dx} (\tan x)$$

$$\Rightarrow \frac{1}{y} \frac{dy}{dx} = \tan x \cdot \frac{1}{\sin x} \cdot \frac{d}{dx} (\sin x) + \log \sin x \cdot \sec^2 x$$

$$\Rightarrow \frac{1}{y} \frac{dy}{dx} = \tan x \cdot \frac{1}{\sin x} \cdot \cos x + \log \sin x \cdot \sec^2 x$$

$$\Rightarrow \frac{1}{y} \frac{dy}{dx} = \tan x \cdot \cot x + \log \sin x \cdot \sec^2 x$$

$$\Rightarrow \frac{dy}{dx} = y [1 + \log \sin x \cdot \sec^2 x]$$

$$\Rightarrow \frac{dy}{dx} = \sin x^{\tan x} [1 + \log \sin x \cdot \sec^2 x]$$

* If $x+y = \sin xy$ then find $\frac{dy}{dx}$.

$$\Rightarrow x+y = \sin xy$$

diff. w.r. to 'x' both side.

$$\Rightarrow \frac{d}{dx} (x+y) = \frac{d}{dx} [\sin xy]$$

$$\Rightarrow 1 + \frac{dy}{dx} = \cos xy \cdot \frac{d}{dx} (xy)$$

$$\Rightarrow 1 + \frac{dy}{dx} = \cos xy \cdot [x \frac{d}{dx} (y) + y \cdot \frac{d}{dx} (x)]$$

$$\Rightarrow 1 + \frac{dy}{dx} = \cos xy \cdot [x \frac{dy}{dx} + y]$$

$$\Rightarrow 1 + \frac{dy}{dx} = \cos xy \cdot x \frac{dy}{dx} + \cos xy \cdot y$$

$$\Rightarrow \frac{dy}{dx} - x \cdot \cos xy \frac{dy}{dx} = y \cdot \cos xy - 1$$

$$\Rightarrow \frac{dy}{dx} [1 - x \cdot \cos xy] = y \cdot \cos xy - 1$$

$$\Rightarrow \frac{dy}{dx} = \frac{y \cdot \cos xy - 1}{1 - x \cdot \cos xy}$$

* If $x^3 + y^3 = 3axy$ then find $\frac{dy}{dx}$.

$$\Rightarrow x^3 + y^3 = 3axy$$

diff. w.r. to 'x' both side.

$$\Rightarrow \frac{d}{dx} (x^3 + y^3) = \frac{d}{dx} (3axy)$$

$$\Rightarrow 3x^2 + 3y^2 \frac{dy}{dx} = 3a \frac{d}{dx} (xy)$$

$$\Rightarrow 3x^2 + 3y^2 \frac{dy}{dx} = 3a [x \cdot \frac{d}{dx} (y) + y \cdot \frac{d}{dx} (x)]$$

$$\Rightarrow 3x^2 + 3y^2 \frac{dy}{dx} = 3a [x \cdot \frac{dy}{dx} + y]$$

$$\Rightarrow 3x^2 + 3y^2 \frac{dy}{dx} = 3ax \frac{dy}{dx} + 3ay$$

$$\Rightarrow 3y^2 \frac{dy}{dx} - 3ax \frac{dy}{dx} = 3ay - 3x^2$$

$$\Rightarrow \frac{dy}{dx} [3y^2 - 3ax] = 3ay - 3x^2$$

$$\Rightarrow \frac{dy}{dx} = \frac{3ay - 3x^2}{3y^2 - 3ax} = \frac{3(ay - x^2)}{3(y^2 - ax)}$$

$$\Rightarrow \frac{dy}{dx} = \frac{ay - x^2}{y^2 - ax}$$

* If $y = \sin(x+y)$ then find $\frac{dy}{dx}$.

$$\Rightarrow y = \sin(x+y)$$

diff. w.r. to 'x' both side.

$$\Rightarrow \frac{d}{dx} (y) = \frac{d}{dx} [\sin(x+y)]$$

$$\Rightarrow \frac{dy}{dx} = \cos(x+y) \cdot \frac{d}{dx} (x+y)$$

$$\Rightarrow \frac{dy}{dx} = \cos(x+y) \cdot [1 + \frac{dy}{dx}]$$

$$\Rightarrow \frac{dy}{dx} = \cos(x+y) + \cos(x+y) \frac{dy}{dx}$$

$$\Rightarrow \frac{dy}{dx} - \cos(x+y) \frac{dy}{dx} = \cos(x+y)$$

$$\Rightarrow \frac{dy}{dx} [1 - \cos(x+y)] = \cos(x+y)$$

$$\Rightarrow \frac{dy}{dx} = \frac{\cos(x+y)}{1 - \cos(x+y)}$$

* If $x \cdot \sin y + y \cdot \sin x = 5$ then find $\frac{dy}{dx}$.

$$\Rightarrow x \cdot \sin y + y \cdot \sin x = 5$$

diff. w.r. to 'x' both side.

$$\Rightarrow \frac{d}{dx} (x \cdot \sin y + y \cdot \sin x) = \frac{d}{dx} (5)$$

$$\Rightarrow x \cdot \frac{d}{dx} (\sin y) + \sin y \cdot \frac{d}{dx} (x) + y \cdot \frac{d}{dx} (\sin x) + \sin x \cdot \frac{d}{dx} (y) = 0$$

$$\Rightarrow x \cdot \cos y \frac{dy}{dx} + \sin y + y \cdot (\cos x + \sin x \cdot \frac{dy}{dx}) = 0$$

$$\Rightarrow x \cdot \cos y \frac{dy}{dx} + \sin y + y \cdot \cos x + y \cdot \sin x \frac{dy}{dx} = -\sin y - y \cdot \cos x$$

$$\Rightarrow \frac{dy}{dx} [x \cdot \cos y + y \cdot \sin x] = -(\sin y + y \cdot \cos x)$$

$$\Rightarrow \frac{dy}{dx} = \frac{-(\sin y + y \cdot \cos x)}{x \cdot \cos y + y \cdot \sin x}$$

* If $x = a(\theta + \sin \theta)$, $y = a(1 - \cos \theta)$ then find $\frac{dy}{dx}$.

$$\Rightarrow x = a(\theta + \sin \theta)$$

diff. w.r. to θ .

$$\Rightarrow \frac{dx}{d\theta} = \frac{d}{d\theta} [a(\theta + \sin \theta)]$$

$$\Rightarrow \frac{dx}{d\theta} = a(1 + \cos \theta)$$

$$y = a(1 - \cos \theta)$$

diff. w.r. to θ .

$$\Rightarrow \frac{dy}{d\theta} = \frac{d}{d\theta} [a(1 - \cos \theta)]$$

$$\Rightarrow \frac{dy}{d\theta} = a \sin \theta$$

$$\text{Now, } \frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$$

$$= \frac{a \sin \theta}{a(1 + \cos \theta)}$$

$$\Rightarrow \frac{dy}{dx} = \frac{\sin \theta}{1 + \cos \theta}$$

$$\Rightarrow \frac{dy}{dx} = \frac{2 \sin \theta/2 \cdot \cos \theta/2}{2 \cos^2 \theta/2}$$

$$\Rightarrow \frac{dy}{dx} = \tan \theta/2$$

* If $x = at^2$, $y = 2at$ then find $\frac{d^2y}{dx^2}$.

$$\Rightarrow x = at^2$$

diff. w.r. to 't'

$$\Rightarrow \frac{dx}{dt} = \frac{d}{dt}(at^2)$$

$$= a(2t)$$

$$\Rightarrow \frac{dx}{dt} = 2at$$

$$y = 2at$$

diff. w.r. to 't'.

$$\Rightarrow \frac{dy}{dt} = \frac{d}{dt}(2at)$$

$$\Rightarrow \frac{dy}{dt} = 2a = 2a.$$

$$\text{Now, } \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$$

$$\Rightarrow \frac{dy}{dx} = \frac{2a}{2at} = \frac{1}{t}$$

Again diff. w.r. to 'x'

$$\Rightarrow \frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{1}{t}\right)$$

$$= \frac{d}{dt}\left(\frac{1}{t}\right) \cdot \frac{dt}{dx}$$

$$= -\frac{1}{t^2} \cdot \frac{1}{2at} \quad (\because \frac{dx}{dt} = 2at)$$

$$\Rightarrow \frac{d^2y}{dx^2} = -\frac{1}{2at^3}$$

* If $x = \sec\theta + \tan\theta$, $y = \sec\theta - \tan\theta$ then find $\frac{dy}{dx}$.

$$\Rightarrow x = \sec\theta + \tan\theta$$

diff. w.r. to 'θ'

$$\Rightarrow \frac{dx}{d\theta} = \frac{d}{d\theta}(\sec\theta + \tan\theta)$$

$$= \sec\theta \cdot \tan\theta + \sec^2\theta$$

$$\Rightarrow \frac{dx}{d\theta} = \sec\theta(\tan\theta + \sec\theta)$$

$$\Rightarrow y = \sec\theta - \tan\theta$$

diff. w.r. to 'θ'

$$\Rightarrow \frac{dy}{d\theta} = \frac{d}{d\theta}(\sec\theta - \tan\theta)$$

$$= \sec\theta \cdot \tan\theta - \sec^2\theta$$

$$\Rightarrow \frac{dy}{d\theta} = \sec\theta(\tan\theta - \sec\theta)$$

$$\text{Now, } \frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$$

$$= \frac{\sec\theta(\tan\theta - \sec\theta)}{\sec\theta(\tan\theta + \sec\theta)}$$

$$= \frac{-(\sec\theta - \tan\theta)}{\sec\theta + \tan\theta}$$

$$\Rightarrow \frac{dy}{dx} = -\frac{y}{x}$$

* If $y = A \cos pt + B \sin pt$ then Prove that $\frac{d^2y}{dt^2} + p^2y = 0$

$$\frac{d^2y}{dt^2} + p^2y = 0$$

$$\Rightarrow y = A \cos pt + B \sin pt$$

$$\Rightarrow \frac{dy}{dt} = \frac{d}{dt}(A \cos pt + B \sin pt)$$

$$= \frac{d}{dt}(A \cos pt) + \frac{d}{dt}(B \sin pt)$$

$$\Rightarrow \frac{dy}{dt} = A(-\sin pt) \cdot \frac{d}{dt}(pt) + B \cos pt \cdot \frac{d}{dt}(pt)$$

$$= -A \sin pt \cdot p + B \cos pt \cdot p$$

$$\Rightarrow \frac{dy}{dt} = p[-A \sin pt + B \cos pt]$$

$$\text{Now } \frac{d^2y}{dt^2} = \frac{d}{dt}[p(-A \sin pt + B \cos pt)]$$

$$= p\left[\frac{d}{dt}(-A \sin pt) + \frac{d}{dt}(B \cos pt)\right]$$

$$= p\left[-A \cos pt \cdot \frac{d}{dt}(pt) + B(-\sin pt) \cdot \frac{d}{dt}(pt)\right]$$

$$= p[-A \cos pt \cdot p - B \sin pt \cdot p]$$

$$= -p^2[A \cos pt + B \sin pt]$$

$$\Rightarrow \frac{d^2y}{dt^2} = -p^2y$$

$$\Rightarrow \frac{d^2y}{dt^2} + p^2y = 0$$

* If $y = e^{2x}$ then prove that $\frac{d^2y}{dx^2} - \frac{dy}{dx} - 2y = 0$

$$\Rightarrow y = e^{2x}$$

$$\Rightarrow \frac{dy}{dx} = \frac{d}{dx}(e^{2x})$$

$$= e^{2x} \cdot \frac{d}{dx}(2x)$$

$$\Rightarrow \frac{dy}{dx} = e^{2x} \cdot 2$$

$$\Rightarrow \frac{d^2y}{dx^2} = \frac{d}{dx}(e^{2x} \cdot 2)$$

$$= 2e^{2x} \cdot \frac{d}{dx}(2x)$$

$$= 2 \cdot e^{2x} \cdot 2$$

$$\Rightarrow \frac{d^2y}{dx^2} = 4e^{2x}$$

$$\text{Now, L.H.S} = \frac{d^2y}{dx^2} - \frac{dy}{dx} - 2y$$

$$= 4e^{2x} - 2e^{2x} - 2e^{2x}$$

$$= 4e^{2x} - 4e^{2x}$$

$$= 0 = \text{R.H.S.}$$

* If $g(x) = x^3 - 3x + 11$ then find maximum and minimum values.

$$g(x) = x^3 - 3x + 11$$

$$\Rightarrow g'(x) = 3x^2 - 3, \quad g''(x) = 6x$$

$$\text{Take } g'(x) = 0$$

$$\Rightarrow 3x^2 - 3 = 0$$

$$\Rightarrow x^2 - 1 = 0$$

$$\Rightarrow x^2 = 1$$

$$\Rightarrow x = \pm 1$$

$$\Rightarrow x = 1, x = -1$$

Put the values of x in $f''(x)$.

$$\Rightarrow \text{For } x = 1$$

$$\Rightarrow f''(1) = 6(1) = 6 > 0$$

$\therefore f''$ is minimum.

$$\begin{aligned} \Rightarrow f(1) &= (1)^3 - 3(1) + 11 \\ &= 1 - 3 + 11 \\ &= 9. \end{aligned}$$

$$\Rightarrow \text{For } x = -1$$

$$\Rightarrow f''(-1) = 6(-1) = -6 < 0$$

$\therefore f''$ is maximum.

$$\begin{aligned} \Rightarrow f(-1) &= (-1)^3 - 3(-1) + 11 \\ &= -1 + 3 + 11 \\ &= 13. \end{aligned}$$

* If $f(x) = 2x^3 - 3x^2 - 12x + 5$ then find Maximum and minimum values.

$$f(x) = 2x^3 - 3x^2 - 12x + 5$$

$$\Rightarrow f'(x) = 6x^2 - 6x - 12, f''(x) = 12x - 6$$

$$\text{Take } f'(x) = 0.$$

$$\Rightarrow 6x^2 - 6x - 12 = 0$$

$$\Rightarrow x^2 - x - 2 = 0$$

$$\Rightarrow (x-2)(x+1) = 0$$

$$\Rightarrow x-2=0 \text{ or } x+1=0.$$

$$\Rightarrow x=2 \text{ or } x=-1$$

Put the values of x in $f''(x)$.

$$\Rightarrow \text{For } x=2$$

$$\begin{aligned} \Rightarrow f''(2) &= 12(2) - 6 \\ &= 24 - 6 \\ &= 18 > 0 \end{aligned}$$

f'' is minimum.

$$\begin{aligned} \Rightarrow f(2) &= 2(2)^3 - 3(2)^2 - 12(2) + 5 \\ &= 2(8) - 3(4) - 24 + 5 \\ &= 16 - 12 - 24 + 5 \\ f(2) &= -15 \end{aligned}$$

$$\Rightarrow \text{For } x=-1$$

$$\begin{aligned} \Rightarrow f''(-1) &= 12(-1) - 6 \\ &= -12 - 6 \\ &= -18 < 0 \end{aligned}$$

f'' is maximum.

$$\begin{aligned} \Rightarrow f(-1) &= 2(-1)^3 - 3(-1)^2 - 12(-1) + 5 \\ &= 2(-1) - 3(1) + 12 + 5 \\ &= -2 - 3 + 12 + 5 \\ f(-1) &= 12. \end{aligned}$$

* If $f(x) = 2x^3 - 15x^2 + 36x + 10$ then find Maximum and minimum values.

$$f(x) = 2x^3 - 15x^2 + 36x + 10.$$

$$\Rightarrow f'(x) = 6x^2 - 30x + 36, f''(x) = 12x - 30.$$

$$\text{Take } f'(x) = 0$$

$$\Rightarrow 6x^2 - 30x + 36 = 0$$

$$\Rightarrow x^2 - 5x + 6 = 0$$

$$\Rightarrow (x-2)(x-3) = 0$$

$$\Rightarrow x-2=0 \text{ or } x-3=0$$

$$\Rightarrow x=2 \text{ or } x=3.$$

$$\Rightarrow \text{For } x=2$$

$$\begin{aligned} \Rightarrow f''(2) &= 12(2) - 30 \\ &= 24 - 30 \\ &= -6 < 0 \end{aligned}$$

f'' is maximum.

$$\begin{aligned} \Rightarrow f(2) &= 2(2)^3 - 15(2)^2 + 36(2) + 10 \\ &= 2(8) - 15(4) + 72 + 10 \\ &= 16 - 60 + 72 + 10 \\ &= 38. \end{aligned}$$

$$\Rightarrow \text{For } x=3$$

f'' is min.

$$\begin{aligned} \Rightarrow f(3) &= 2(3)^3 - 15(3)^2 + 36(3) + 10 \\ &= 2(27) - 15(9) + 108 + 10 \\ &= 54 - 135 + 108 + 10 \\ &= 37. \end{aligned}$$

* If $f(x) = x + \frac{1}{x}$ then find maximum and minimum values.

$$f(x) = x + \frac{1}{x}$$

$$\Rightarrow f'(x) = 1 - \frac{1}{x^2}, f''(x) = -\left(\frac{-2}{x^3}\right) = \frac{2}{x^3}.$$

$$\Rightarrow \text{Take } f'(x) = 0$$

$$\Rightarrow 1 - \frac{1}{x^2} = 0$$

$$\Rightarrow 1 = \frac{1}{x^2}$$

$$\Rightarrow x^2 = 1$$

$$\Rightarrow x = \pm 1$$

$$\Rightarrow x=1, x=-1.$$

Put the values of x in $f''(x)$

$$\Rightarrow \text{For } x=1$$

$$\begin{aligned} \Rightarrow f''(1) &= \frac{2}{(1)^3} \\ &= 2 > 0 \end{aligned}$$

f'' is minimum.

$$\begin{aligned} \Rightarrow f(1) &= 1 + \frac{1}{1} \\ &= 1 + 1 \\ &= 2. \end{aligned}$$

$$\Rightarrow \text{For } x=-1$$

$$\begin{aligned} \Rightarrow f''(-1) &= \frac{2}{(-1)^3} \\ &= -2 < 0 \end{aligned}$$

f'' is maximum

$$\begin{aligned} \Rightarrow f(-1) &= -1 + \frac{1}{-1} \\ &= -1 - 1 \\ &= -2. \end{aligned}$$

* The equation of motion of particle is $s = t^3 - 3t^2 + 4t + 3$ then find velocity and acceleration at $t=2$.

$$\Rightarrow s = t^3 - 3t^2 + 4t + 3$$

$$v = \frac{ds}{dt} = 3t^2 - 6t + 4$$

$$a = \frac{dv}{dt} = 6t - 6$$

at $t=2$.

$$\Rightarrow V = 3(2)^2 - 6(2) + 4$$

$$= 12 - 12 + 4$$

$$\boxed{V = 4 \text{ units}}$$

$$\Rightarrow a = 6(2) - 6$$

$$= 12 - 6$$

$$\boxed{a = 6 \text{ units}}$$

* The equation of motion of a Particle is $S = t^3 - 6t^2 + 9t + 6$. Find V when $a=0$.

$$S = t^3 - 6t^2 + 9t + 6$$

$$V = \frac{dS}{dt} = 3t^2 - 12t + 9$$

$$a = \frac{dV}{dt} = 6t - 12$$

$$\Rightarrow \text{Now } a = 0$$

$$\Rightarrow 6t - 12 = 0$$

$$\Rightarrow 6t = 12$$

$$\Rightarrow t = \frac{12}{6} = 2 \text{ sec.}$$

Put $t=2$ in velocity.

$$\Rightarrow V = 3(2)^2 - 12(2) + 9$$

$$= 12 - 24 + 9$$

$$\boxed{V = -3 \text{ units}}$$

* The equation of motion of Particle is $S = t^3 + 3t$ then.

(i) Find velocity and acceleration at $t=3$.

(ii) when velocity and acceleration will become equal?

$$S = t^3 + 3t$$

$$\Rightarrow V = \frac{dS}{dt} = 3t^2 + 3$$

$$a = \frac{dV}{dt} = 6t$$

(i) V and a at $t=3$.

$$\Rightarrow V = 3(3)^2 + 3$$

$$= 27 + 3$$

$$\boxed{V = 30 \text{ units}}$$

$$\Rightarrow a = 6(3)$$

$$\boxed{a = 18 \text{ units}}$$

(ii) velocity and acceleration are equal then $V = a$

$$\Rightarrow 3t^2 + 3 = 6t$$

$$\Rightarrow 3t^2 - 6t + 3 = 0$$

$$\Rightarrow t^2 - 2t + 1 = 0$$

$$\Rightarrow (t-1)^2 = 0$$

$$\Rightarrow t-1 = 0$$

$$\Rightarrow \boxed{t = 1 \text{ sec}}$$

* The equation of motion of a Particle is $S = t^3 - 6t^2 + 9t + 6$ then

(i) Find V and a at $t=2$.

(ii) Find S and a when Particle change its direction.

$$S = t^3 - 6t^2 + 9t + 6$$

$$V = \frac{dS}{dt} = 3t^2 - 12t + 9$$

$$a = \frac{dV}{dt} = 6t - 12$$

(i) V and a at $t=2$.

$$\Rightarrow V = 3(2)^2 - 12(2) + 9$$

$$= 12 - 24 + 9$$

$$\boxed{V = -3 \text{ units}}$$

$$\Rightarrow a = 6(2) - 12$$

$$= 12 - 12$$

$$\boxed{a = 0 \text{ unit}}$$

(ii) when Particle change its direction

$$V = 0$$

$$\Rightarrow 3t^2 - 12t + 9 = 0$$

$$\Rightarrow t^2 - 4t + 3 = 0$$

$$\Rightarrow (t-3)(t-1) = 0$$

$$\Rightarrow t-3 = 0 \quad \text{or} \quad t-1 = 0$$

$$\Rightarrow t = 3 \quad \text{or} \quad t = 1$$

$\Rightarrow$ at $t=1$

$$\Rightarrow S = (1)^3 - 6(1)^2 + 9(1) + 6$$

$$= 1 - 6 + 9 + 6$$

$$\boxed{S = 10 \text{ units}}$$

$$\Rightarrow a = 6(1) - 12$$

$$= 6 - 12$$

$$\boxed{a = -6 \text{ units}}$$

at $t=3$

$$\Rightarrow S = (3)^3 - 6(3)^2 + 9(3) + 6$$

$$= 27 - 54 + 27 + 6$$

$$\boxed{S = 6 \text{ units}}$$

$$\Rightarrow a = 6(3) - 12$$

$$= 18 - 12$$

$$\boxed{a = 6 \text{ units}}$$

$$\begin{aligned}
 * \text{ Evaluate: } & \int \left(x - \frac{1}{x}\right)^2 dx \\
 &= \int \left(x^2 - 2x \cdot \frac{1}{x} + \frac{1}{x^2}\right) dx \\
 &= \int \left(x^2 - 2 + \frac{1}{x^2}\right) dx \\
 &= \int x^2 dx - 2 \int 1 dx + \int \frac{1}{x^2} dx \\
 &= \frac{x^3}{3} - 2x - \frac{1}{x} + C.
 \end{aligned}$$

$$\begin{aligned}
 * \text{ Evaluate: } & \int \frac{\operatorname{cosec}^2 x}{\sec x} dx \\
 &= \int \operatorname{cosec}^2 x \cdot \frac{1}{\sec x} dx \\
 &= \int \frac{1}{\sin^2 x} \cdot \cos x dx \\
 &= \int \frac{\cos x}{\sin x} \cdot \frac{1}{\sin x} dx \\
 &= \int \cot x \cdot \operatorname{cosec} x dx \\
 &= -\operatorname{cosec} x + C.
 \end{aligned}$$

$$\begin{aligned}
 * \text{ Evaluate: } & \int \frac{\cos 2x}{\cos^2 x \cdot \sin^2 x} dx. \\
 &= \int \frac{\cos^2 x - \sin^2 x}{\cos^2 x \cdot \sin^2 x} dx \quad (\because \cos 2\theta = \cos^2 \theta - \sin^2 \theta) \\
 &= \int \left(\frac{\cos^2 x}{\cos^2 x \cdot \sin^2 x} - \frac{\sin^2 x}{\cos^2 x \cdot \sin^2 x} \right) dx \\
 &= \int \left(\frac{1}{\sin^2 x} - \frac{1}{\cos^2 x} \right) dx \\
 &= \int \operatorname{cosec}^2 x dx - \int \sec^2 x dx \\
 &= -\cot x - \tan x + C.
 \end{aligned}$$

$$\begin{aligned}
 * \text{ Evaluate: } & \int \frac{2 + 3 \sin x}{\cos^2 x} dx. \\
 &= \int \left(\frac{2}{\cos^2 x} + \frac{3 \sin x}{\cos^2 x} \right) dx \\
 &= 2 \int \frac{1}{\cos^2 x} dx + 3 \int \frac{\sin x}{\cos^2 x} \cdot \frac{1}{\cos x} dx \\
 &= 2 \int \sec^2 x dx + 3 \int \tan x \cdot \sec x dx \\
 &= 2 \tan x + 3 \sec x + C.
 \end{aligned}$$

$$\begin{aligned}
 * \text{ Evaluate: } & \int \frac{4 + 3 \cos x}{\sin^2 x} dx. \\
 &= \int \left(\frac{4}{\sin^2 x} + \frac{3 \cos x}{\sin^2 x} \right) dx \\
 &= 4 \int \frac{1}{\sin^2 x} dx + 3 \int \frac{\cos x}{\sin x} \cdot \frac{1}{\sin x} dx \\
 &= 4 \int \operatorname{cosec}^2 x dx + 3 \int \cot x \cdot \operatorname{cosec} x dx \\
 &= 4(-\cot x) + 3(-\operatorname{cosec} x) + C \\
 &= -4 \cot x - 3 \operatorname{cosec} x + C.
 \end{aligned}$$

$$\begin{aligned}
 * \text{ Evaluate: } & \int \frac{\sin^3 x + \cos^3 x}{\sin^2 x \cdot \cos^2 x} dx. \\
 &= \int \left(\frac{\sin^3 x}{\sin^2 x \cdot \cos^2 x} + \frac{\cos^3 x}{\sin^2 x \cdot \cos^2 x} \right) dx \\
 &= \int \left(\frac{\sin x}{\cos^2 x} + \frac{\cos x}{\sin^2 x} \right) dx \\
 &= \int \frac{\sin x}{\cos x} \cdot \frac{1}{\cos x} dx + \int \frac{\cos x}{\sin x} \cdot \frac{1}{\sin x} dx \\
 &= \int \tan x \cdot \sec x dx + \int \cot x \cdot \operatorname{cosec} x dx \\
 &= \sec x - \operatorname{cosec} x + C.
 \end{aligned}$$

$$\begin{aligned}
 * \text{ Evaluate: } & \int \frac{\cos x}{1 + \cos x} dx. \\
 &= \int \frac{\cos x}{1 + \cos x} \times \frac{1 - \cos x}{1 - \cos x} dx \\
 &= \int \frac{\cos x (1 - \cos x)}{1 - \cos^2 x} dx \\
 &= \int \frac{\cos x - \cos^2 x}{\sin^2 x} dx \\
 &= \int \left(\frac{\cos x}{\sin^2 x} - \frac{\cos^2 x}{\sin^2 x} \right) dx \\
 &= \int \left(\frac{\cos x}{\sin x} \cdot \frac{1}{\sin x} - \cot^2 x \right) dx.
 \end{aligned}$$

$$= \int (\operatorname{cosec} x \cdot \cot x - (\operatorname{cosec}^2 x - 1)) dx$$

$$= \int (\operatorname{cosec} x \cdot \cot x - \operatorname{cosec}^2 x + 1) dx$$

$$= \int \operatorname{cosec} x \cdot \cot x dx - \int \operatorname{cosec}^2 x dx + \int 1 dx$$

$$= -\operatorname{cosec} x + \cot x + x + C.$$

* Evaluate: $\int x \cdot e^x dx$

$$\Rightarrow u = x \quad \left| \quad v = e^x \right.$$

$$\frac{du}{dx} = \frac{d}{dx}(x) = 1 \quad \left| \quad \int v dx = \int e^x dx = e^x \right.$$

$$\int u \cdot v dx = u \cdot \int v dx - \int \left(\frac{du}{dx} \cdot \int v dx \right) dx$$

$$= x \cdot e^x - \int (1 \cdot e^x) dx$$

$$= x \cdot e^x - \int e^x dx$$

$$= x \cdot e^x - e^x + C$$

$$= e^x(x-1) + C.$$

* Evaluate: $\int \sin x \cdot x dx$

$$\Rightarrow u = x \quad \left| \quad v = \sin x \right.$$

$$\Rightarrow \frac{du}{dx} = \frac{d}{dx}(x) = 1 \quad \left| \quad \int v dx = \int \sin x dx \right.$$

$$\quad \quad \quad = -\cos x$$

$$\int u \cdot v dx = u \cdot \int v dx - \int \left(\frac{du}{dx} \cdot \int v dx \right) dx$$

$$= x \cdot (-\cos x) - \int (1 \cdot (-\cos x)) dx$$

$$= -x \cdot \cos x - \int (-\cos x) dx$$

$$= -x \cdot \cos x + \int \cos x dx$$

$$= -x \cdot \cos x + \sin x + C.$$

* Evaluate: $\int x^2 \cdot \log x dx$

$$\Rightarrow u = \log x \quad \left| \quad v = x^2 \right.$$

$$\Rightarrow \frac{du}{dx} = \frac{d}{dx}(\log x) \quad \left| \quad \int v dx = \int x^2 dx \right.$$

$$= \frac{1}{x} \quad \left| \quad = \frac{x^3}{3} \right.$$

$$\int u \cdot v dx = u \cdot \int v dx - \int \left(\frac{du}{dx} \cdot \int v dx \right) dx$$

$$= \log x \cdot \frac{x^3}{3} - \int \left(\frac{1}{x} \cdot \frac{x^3}{3} \right) dx$$

$$= \log x \cdot \frac{x^3}{3} - \int \frac{x^2}{3} dx$$

$$= \log x \cdot \frac{x^3}{3} - \frac{1}{3} \int x^2 dx$$

$$= \log x \cdot \frac{x^3}{3} - \frac{1}{3} \cdot \frac{x^3}{3} + C$$

$$= \frac{x^3}{3} \left(\log x - \frac{1}{3} \right) + C.$$

* Evaluate: $\int x^2 \cdot e^x dx$

$$\Rightarrow u = x^2 \quad \left| \quad v = e^x \right.$$

$$\Rightarrow \frac{du}{dx} = \frac{d}{dx}(x^2) \quad \left| \quad \int v dx = \int e^x dx \right.$$

$$= 2x \quad \left| \quad = e^x \right.$$

$$\int u \cdot v dx = u \cdot \int v dx - \int \left(\frac{du}{dx} \cdot \int v dx \right) dx$$

$$= x^2 \cdot e^x - \int (2x \cdot e^x) dx$$

$$= x^2 \cdot e^x - 2 \int x \cdot e^x dx$$

$$= x^2 \cdot e^x - 2 \left[x \cdot \int e^x - \int \left(\frac{d}{dx}(x) \cdot \int e^x dx \right) \right]$$

$$= x^2 \cdot e^x - 2 \left[x \cdot e^x - \int (1 \cdot e^x) dx \right]$$

$$= x^2 \cdot e^x - 2 \left[x \cdot e^x - \int e^x dx \right]$$

$$= x^2 \cdot e^x - 2 \left[x \cdot e^x - e^x \right] + C$$

$$= x^2 \cdot e^x - 2 \left[e^x(x-1) \right] + C$$

$$= x^2 \cdot e^x - 2e^x(x-1) + C.$$

* Evaluate: $\int x \cdot e^{3x} dx$

$$\Rightarrow u = x \quad \left| \quad v = e^{3x} \right.$$

$$\Rightarrow \frac{du}{dx} = \frac{d}{dx}(x)$$

$$= 1 \quad \left| \quad \int v dx = \int e^{3x} dx \right.$$

$$= \frac{e^{3x}}{3}$$

$$\int u \cdot v dx = u \cdot \int v dx - \int \left(\frac{du}{dx} \cdot \int v dx \right) dx$$

$$= x \cdot \frac{e^{3x}}{3} - \int \left(1 \cdot \frac{e^{3x}}{3} \right) dx$$

$$= x \cdot \frac{e^{3x}}{3} - \int \frac{e^{3x}}{3} dx$$

$$= x \cdot \frac{e^{3x}}{3} - \frac{1}{3} \int e^{3x} dx$$

$$= x \cdot \frac{e^{3x}}{3} - \frac{1}{3} \frac{e^{3x}}{3} + C$$

$$= \frac{e^{3x}}{3} \left(x - \frac{1}{3} \right) + C.$$

* Evaluate: $\int e^{\tan x} \cdot \sec^2 x dx$

Suppose $u = \tan x$

$$\Rightarrow \frac{du}{dx} = \frac{d}{dx}(\tan x)$$

$$\Rightarrow \frac{du}{dx} = \sec^2 x$$

$$\Rightarrow du = \sec^2 x dx$$

Now $\int e^{\tan x} \cdot \sec^2 x dx$

$$= \int e^u \cdot du$$

$$= e^u + C$$

$$= e^{\tan x} + C.$$

Evaluate:

* $\int \cos x \cdot \sqrt{\sin x} dx$

Suppose $u = \sin x$

$$\Rightarrow \frac{du}{dx} = \frac{d}{dx}(\sin x)$$

$$\Rightarrow \frac{du}{dx} = \cos x$$

$$\Rightarrow du = \cos x dx$$

Now, $\int \cos x \cdot \sqrt{\sin x} dx$

$$= \int \sqrt{\sin x} \cdot \cos x dx$$

$$= \int \sqrt{u} \cdot du$$

$$= \int (u)^{1/2} du$$

$$= \frac{u^{1/2+1}}{\frac{1}{2}+1} + C$$

$$= \frac{u^{3/2}}{3/2} + C$$

$$= \frac{(\sin x)^{3/2}}{3/2} + C.$$

* Evaluate: $\int \frac{\sin(\log x)}{x} dx$

Suppose $u = \log x$

$$\Rightarrow \frac{du}{dx} = \frac{d}{dx}(\log x)$$

$$\Rightarrow \frac{du}{dx} = \frac{1}{x}$$

$$\Rightarrow du = \frac{1}{x} dx$$

Now, $\int \frac{\sin(\log x)}{x} dx$

$$= \int \sin u \cdot du$$

$$= -\cos(u) + C$$

$$= -\cos(\log x) + C.$$

* Evaluate: $\int \frac{(\log x)^5}{x} dx$

Suppose $\boxed{u = \log x}$

$$\Rightarrow \frac{du}{dx} = \frac{d}{dx}(\log x)$$

$$\Rightarrow \frac{du}{dx} = \frac{1}{x}$$

$$\Rightarrow \boxed{du = \frac{1}{x} dx}$$

Now, $\int \frac{(\log x)^5}{x} dx$

$$= \int (u)^5 \cdot du$$

$$= \frac{u^6}{6} + C$$

$$= \frac{(\log x)^6}{6} + C.$$

* Evaluate: $\int \frac{e^x(x+1)}{\sin^2(x \cdot e^x)} dx$

Suppose $\boxed{u = x \cdot e^x}$

$$\Rightarrow \frac{du}{dx} = \frac{d}{dx}(x \cdot e^x)$$

$$= x \cdot \frac{d}{dx}(e^x) + e^x \cdot \frac{d}{dx}(x)$$

$$= x \cdot e^x + e^x$$

$$\Rightarrow \frac{du}{dx} = e^x(x+1)$$

$$\Rightarrow \boxed{du = e^x(x+1) dx}$$

Now, $\int \frac{e^x(x+1)}{\sin^2(x \cdot e^x)} dx$

$$= \int \frac{du}{\sin^2 u}$$

$$= \int \frac{1}{\sin^2 u} \cdot du$$

$$= \int \operatorname{cosec}^2 u \, du$$

$$= -\cot(u) + C$$

$$= -\cot(x \cdot e^x) + C.$$

* Evaluate: $\int \frac{e^x(x+1)}{\cos^2(x \cdot e^x)} dx$

Suppose $\boxed{u = x \cdot e^x}$

$$\Rightarrow \frac{du}{dx} = \frac{d}{dx}(x \cdot e^x)$$

$$= x \cdot \frac{d}{dx}(e^x) + e^x \cdot \frac{d}{dx}(x)$$

$$= x \cdot e^x + e^x$$

$$\Rightarrow \frac{du}{dx} = e^x(x+1)$$

$$\Rightarrow \boxed{du = e^x(x+1) dx}$$

Now, $\int \frac{e^x(x+1)}{\cos^2(x \cdot e^x)} dx$

$$= \int \frac{du}{\cos^2 u}$$

$$= \int \frac{1}{\cos^2 u} \cdot du$$

$$= \int \sec^2 u \, du$$

$$= \tan u + C$$

$$= \tan(x \cdot e^x) + C.$$

* Evaluate: $\int \frac{2 \tan^7 x}{1+x^2} dx$

Suppose $\boxed{u = \tan^{-1} x}$

$$\Rightarrow \frac{du}{dx} = \frac{d}{dx}(\tan^{-1} x)$$

$$= \frac{1}{1+x^2}$$

$$\Rightarrow \boxed{du = \frac{1}{1+x^2} dx}$$

Now, $\int \frac{2 \tan^7 x}{1+x^2} dx$

$$= \int 2 \cdot u \, du$$

$$= 2 \int u \, du$$

$$= 2 \cdot \frac{u^2}{2} + C$$

$$= u^2 + C$$

$$= (\tan^{-1} x)^2 + C.$$

$$* \int 2x \cdot (x^2+8)^8 dx$$

Suppose $u = x^2+8$

$$\Rightarrow \frac{du}{dx} = \frac{d}{dx}(x^2+8)$$

$$\Rightarrow \frac{du}{dx} = 2x$$

$$\Rightarrow du = 2x \cdot dx$$

Now, $\int 2x \cdot (x^2+8)^8 dx$

$$= \int (x^2+8)^8 \cdot 2x dx$$

$$= \int u^8 \cdot du$$

$$= \frac{u^9}{9} + C$$

$$= \frac{(x^2+8)^9}{9} + C.$$

* Evaluate: $\int_0^1 \frac{x^3-8}{x-2} dx$

$$= \int_0^1 \frac{(x/2)(x^2+2x+4)}{(x/2)} dx$$

$$= \int_0^1 (x^2+2x+4) dx$$

$$= \left[\frac{x^3}{3} + 2 \cdot \frac{x^2}{2} + 4x \right]_0^1$$

$$= \left[\frac{x^3}{3} + x^2 + 4x \right]_0^1$$

$$= \left[\frac{(1)^3}{3} + (1)^2 + 4(1) \right] - \left[\frac{(0)^3}{3} + (0)^2 + 4(0) \right]$$

$$= \frac{1}{3} + 1 + 4 - 0$$

$$= \frac{1+3+12}{3}$$

$$= \frac{16}{3}.$$

$$* \int_1^e \frac{(\log x)^2}{x} dx$$

Suppose $u = \log x$

$$\Rightarrow \frac{du}{dx} = \frac{1}{x}$$

$$\Rightarrow du = \frac{1}{x} dx$$

Now, $\int_1^e \frac{(\log x)^2}{x} dx$

$$= \int_1^e (u)^2 du$$

$$= \left[\frac{u^3}{3} \right]_1^e$$

$$= \left[\frac{(\log x)^3}{3} \right]_1^e$$

$$= \frac{(\log e)^3}{3} - \frac{(\log 1)^3}{3}$$

$$= \frac{(1)^3}{3} - \frac{0}{3}$$

$$= \frac{1}{3}.$$

Evaluate:

* $\int_1^3 (x^2+x+1) dx$

$$= \left[\frac{x^3}{3} + \frac{x^2}{2} + x \right]_1^3$$

$$= \left[\frac{(3)^3}{3} + \frac{(3)^2}{2} + 3 \right] - \left[\frac{(1)^3}{3} + \frac{(1)^2}{2} + 1 \right]$$

$$= \left[\frac{27}{3} + \frac{9}{2} + 3 \right] - \left[\frac{1}{3} + \frac{1}{2} + 1 \right]$$

$$= \left[\frac{54+27+18}{6} \right] - \left[\frac{2+3+6}{6} \right]$$

$$= \left[\frac{99}{6} \right] - \left[\frac{11}{6} \right]$$

$$= \frac{99-11}{6} = \frac{88}{6}.$$

$$\begin{aligned}
 * \text{ Evaluate: } & \int_0^1 \frac{x}{x+1} dx \\
 &= \int_0^1 \frac{x+1-1}{x+1} dx \\
 &= \int_0^1 \left(\frac{x+1}{x+1} - \frac{1}{x+1} \right) dx \\
 &= \int_0^1 \left(1 - \frac{1}{x+1} \right) dx \\
 &= \left[x - \log(x+1) \right]_0^1 \\
 &= [1 - \log(1+1)] - [0 - \log(0+1)] \\
 &= 1 - \log 2 + \log 1 \\
 &= 1 - \log 2.
 \end{aligned}$$

$$\begin{aligned}
 * \text{ Evaluate: } & \int_{-4}^{-3} \frac{x}{7+x} dx \\
 &= \int_{-4}^{-3} \frac{7+x-7}{7+x} dx \\
 &= \int_{-4}^{-3} \left(\frac{7+x}{7+x} - \frac{7}{7+x} \right) dx \\
 &= \int_{-4}^{-3} \left(1 - \frac{7}{7+x} \right) dx \\
 &= \left[x - 7 \log(7+x) \right]_{-4}^{-3} \\
 &= [-3 - 7 \log(7-3)] - [-4 - 7 \log(7-4)] \\
 &= -3 - 7 \log 4 + 4 + 7 \log 3 \\
 &= 1 + 7 \log 3 - 7 \log 4 \\
 &= 1 + 7 (\log 3 - \log 4) \\
 &= 1 + 7 \log \left(\frac{3}{4} \right)
 \end{aligned}$$

$$\begin{aligned}
 * \int_0^1 \frac{2}{1+x^2} dx \\
 &= 2 \int_0^1 \frac{1}{1+x^2} dx \\
 &= 2 \left[\tan^{-1} x \right]_0^1 \\
 &= 2 \left[\tan^{-1}(1) - \tan^{-1}(0) \right] \\
 &= 2 \left[\frac{\pi}{4} - 0 \right] \\
 &= 2 \times \frac{\pi}{4} \\
 &= \frac{\pi}{2}.
 \end{aligned}$$

$$* \text{ Evaluate: } \int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx$$

$$I = \int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx \quad \text{--- (1)}$$

$$= \int_0^{\pi/2} \frac{\sqrt{\sin(\frac{\pi}{2}-x)}}{\sqrt{\sin(\frac{\pi}{2}-x)} + \sqrt{\cos(\frac{\pi}{2}-x)}} dx$$

$$I = \int_0^{\pi/2} \frac{\sqrt{\cos x}}{\sqrt{\cos x} + \sqrt{\sin x}} dx \quad \text{--- (2)}$$

Adding (1) & (2)

$$\Rightarrow 2I = \int_0^{\pi/2} \left(\frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} + \frac{\sqrt{\cos x}}{\sqrt{\cos x} + \sqrt{\sin x}} \right) dx$$

$$= \int_0^{\pi/2} \left(\frac{\sqrt{\sin x} + \sqrt{\cos x}}{\sqrt{\sin x} + \sqrt{\cos x}} \right) dx$$

$$= \int_0^{\pi/2} 1 dx$$

$$= [x]_0^{\pi/2}$$

$$\Rightarrow 2I = \frac{\pi}{2} - 0$$

$$\Rightarrow I = \frac{\pi}{2 \times 2} = \frac{\pi}{4}.$$

* Evaluate: $\int_0^{\pi/2} \frac{\tan x}{\tan x + \cot x} dx$

$$I = \int_0^{\pi/2} \frac{\tan x}{\tan x + \cot x} dx \quad (1)$$

$$= \int_0^{\pi/2} \frac{\tan(\frac{\pi}{2} - x)}{\tan(\frac{\pi}{2} - x) + \cot(\frac{\pi}{2} - x)} dx$$

$$I = \int_0^{\pi/2} \frac{\cot x}{\cot x + \tan x} dx \quad (2)$$

Adding (1) & (2)

$$\Rightarrow 2I = \int_0^{\pi/2} \left(\frac{\tan x}{\tan x + \cot x} + \frac{\cot x}{\cot x + \tan x} \right) dx$$

$$= \int_0^{\pi/2} \left(\frac{\tan x + \cot x}{\tan x + \cot x} \right) dx$$

$$= \int_0^{\pi/2} 1 dx$$

$$= [x]_0^{\pi/2}$$

$$\Rightarrow 2I = \frac{\pi}{2} - 0$$

$$\Rightarrow I = \frac{\pi}{4}$$

* Evaluate: $\int_0^{\pi/2} \frac{\sqrt{\sec x}}{\sqrt{\sec x} + \sqrt{\cos x}} dx$

$$I = \int_0^{\pi/2} \frac{\sqrt{\sec x}}{\sqrt{\sec x} + \sqrt{\cos x}} dx \quad (1)$$

$$= \int_0^{\pi/2} \frac{\sqrt{\sec(\frac{\pi}{2} - x)}}{\sqrt{\sec(\frac{\pi}{2} - x)} + \sqrt{\cos(\frac{\pi}{2} - x)}} dx$$

$$= \int_0^{\pi/2} \frac{\sqrt{\cos x}}{\sqrt{\cos x} + \sqrt{\sec x}} dx \quad (2)$$

Adding (1) & (2)

$$\Rightarrow 2I = \int_0^{\pi/2} \left(\frac{\sqrt{\sec x}}{\sqrt{\sec x} + \sqrt{\cos x}} + \frac{\sqrt{\cos x}}{\sqrt{\cos x} + \sqrt{\sec x}} \right) dx$$

$$= \int_0^{\pi/2} \left(\frac{\sqrt{\sec x} + \sqrt{\cos x}}{\sqrt{\sec x} + \sqrt{\cos x}} \right) dx$$

$$= \int_0^{\pi/2} 1 dx$$

$$= [x]_0^{\pi/2}$$

$$\Rightarrow 2I = \frac{\pi}{2} - 0$$

$$\Rightarrow I = \frac{\pi}{4}$$

* $\int_0^{\pi/2} \frac{1}{1 + \sqrt{\tan x}} dx$

$$= \int_0^{\pi/2} \frac{1}{1 + \frac{\sqrt{\sin x}}{\sqrt{\cos x}}} dx$$

$$= \int_0^{\pi/2} \frac{1}{\frac{\sqrt{\cos x} + \sqrt{\sin x}}{\sqrt{\cos x}}} dx$$

$$I = \int_0^{\pi/2} \frac{\sqrt{\cos x}}{\sqrt{\cos x} + \sqrt{\sin x}} dx \quad (1)$$

$$= \int_0^{\pi/2} \frac{\sqrt{\cos(\frac{\pi}{2} - x)}}{\sqrt{\cos(\frac{\pi}{2} - x)} + \sqrt{\sin(\frac{\pi}{2} - x)}} dx$$

$$I = \int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx \quad (2)$$

Adding (1) & (2)

$$\Rightarrow 2I = \int_0^{\pi/2} \left(\frac{\sqrt{\cos x}}{\sqrt{\cos x} + \sqrt{\sin x}} + \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} \right) dx$$

$$= \int_0^{\pi/2} \left(\frac{\sqrt{\cos x} + \sqrt{\sin x}}{\sqrt{\cos x} + \sqrt{\sin x}} \right) dx$$

$$= \int_0^{\pi/2} 1 dx$$

$$= [x]_0^{\pi/2}$$

$$\Rightarrow 2I = \frac{\pi}{2} - 0$$

$$\Rightarrow I = \frac{\pi}{4}$$

Evaluate:

$$* \int_0^{\pi/2} \log(\tan x) dx$$

$$I = \int_0^{\pi/2} \log(\tan x) dx \quad \text{--- (1)}$$

$$= \int_0^{\pi/2} \log[\tan(\frac{\pi}{2} - x)] dx$$

$$I = \int_0^{\pi/2} \log(\cot x) dx \quad \text{--- (2)}$$

Adding (1) & (2)

$$\Rightarrow 2I = \int_0^{\pi/2} [\log(\tan x) + \log(\cot x)] dx$$

$$= \int_0^{\pi/2} \log(\tan x \cdot \cot x) dx$$

$$= \int_0^{\pi/2} \log 1 dx$$

$$= \int_0^{\pi/2} 0 dx$$

$$\Rightarrow 2I = 0$$

$$\Rightarrow 2I = 0.$$

* Evaluate: $\int_0^7 \frac{\sqrt{7-x}}{\sqrt{x} + \sqrt{7-x}} dx$

$$I = \int_0^7 \frac{\sqrt{7-x}}{\sqrt{x} + \sqrt{7-x}} dx \quad \text{--- (1)}$$

$$= \int_0^7 \frac{\sqrt{7-(7-x)}}{\sqrt{7-x} + \sqrt{7-(7-x)}} dx$$

$$= \int_0^7 \frac{\sqrt{7-7+x}}{\sqrt{7-x} + \sqrt{7-7+x}} dx$$

$$I = \int_0^7 \frac{\sqrt{x}}{\sqrt{7-x} + \sqrt{x}} dx \quad \text{--- (2)}$$

Adding (1) & (2)

$$\Rightarrow 2I = \int_0^7 \left(\frac{\sqrt{7-x}}{\sqrt{x} + \sqrt{7-x}} + \frac{\sqrt{x}}{\sqrt{7-x} + \sqrt{x}} \right) dx$$

$$= \int_0^7 \left(\frac{\sqrt{7-x} + \sqrt{x}}{\sqrt{7-x} + \sqrt{x}} \right) dx$$

$$= \int_0^7 1 dx$$

$$= [x]_0^7$$

$$\Rightarrow 2I = 7 - 0$$

$$\Rightarrow I = \frac{7}{2}.$$

* Find the area bounded by the curve $x^2 + y^2 = a^2$.

$$\Rightarrow \text{Here, } x^2 + y^2 = a^2$$

$$\Rightarrow y^2 = a^2 - x^2$$

$$\Rightarrow y = \sqrt{a^2 - x^2}$$

$$A = \int_a^b y dx$$

$$= \int_0^a \sqrt{a^2 - x^2} dx$$

$$= \left[\frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1}\left(\frac{x}{a}\right) \right]_0^a$$

$$= \left[\frac{a}{2} \sqrt{a^2 - a^2} + \frac{a^2}{2} \sin^{-1}\left(\frac{a}{a}\right) \right] - \left[\frac{0}{2} \sqrt{a^2 - 0} + \frac{a^2}{2} \sin^{-1}\left(\frac{0}{a}\right) \right]$$

$$= \left[0 + \frac{a^2}{2} \sin^{-1}(1) \right] - \left[0 + \frac{a^2}{2} \sin^{-1}(0) \right]$$

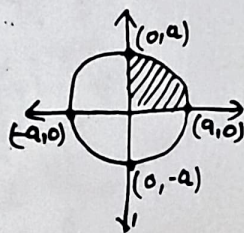
$$= \frac{a^2}{2} \cdot \frac{\pi}{2} - \frac{a^2}{2} \cdot (0)$$

$$A = \frac{\pi a^2}{4}$$

$$\text{Total Area of circle} = 4 \times A$$

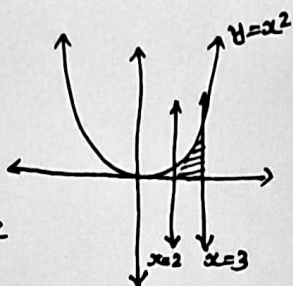
$$= 4 \cdot \frac{\pi a^2}{4}$$

$$= \pi a^2.$$



* Find the area bounded by the curve $y=x^2$ and lines $x=2, x=3$ with x-axis.

$\Rightarrow$ Here, $y=x^2$
 $x=2, x=3$



$$\text{Area } A = \int_a^b y \, dx$$

$$= \int_2^3 x^2 \, dx$$

$$= \left[\frac{x^3}{3} \right]_2^3$$

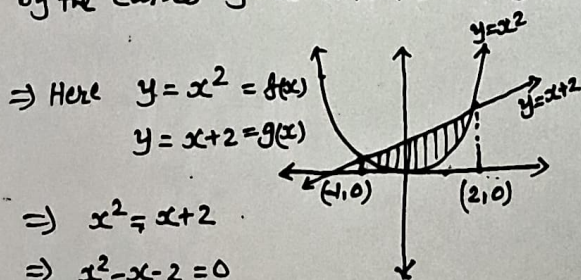
$$= \frac{(3)^3}{3} - \frac{(2)^3}{3}$$

$$= \frac{27}{3} - \frac{8}{3}$$

$$= \frac{27-8}{3}$$

$$= \frac{19}{3} \text{ units.}$$

* Find the area of the region bounded by the curves $y=x^2$ and the line $y=x+2$



$\Rightarrow$ Here $y=x^2 = f(x)$
 $y=x+2 = g(x)$

$$\Rightarrow x^2 = x+2$$

$$\Rightarrow x^2 - x - 2 = 0$$

$$\Rightarrow (x-2)(x+1) = 0$$

$$\Rightarrow x-2=0 \text{ or } x+1=0$$

$$\Rightarrow x=2 \text{ or } x=-1.$$

$$A = \int_a^b (f(x) - g(x)) \, dx$$

$$= \int_{-1}^2 (x^2 - (x+2)) \, dx$$

$$= \int_{-1}^2 (x^2 - x - 2) \, dx$$

$$= \left[\frac{x^3}{3} - \frac{x^2}{2} - 2x \right]_{-1}^2$$

$$= \left[\frac{(2)^3}{3} - \frac{(2)^2}{2} - 2(2) \right] - \left[\frac{(-1)^3}{3} - \frac{(-1)^2}{2} - 2(-1) \right]$$

$$= \left[\frac{8}{3} - \frac{4}{2} - 4 \right] - \left[-\frac{1}{3} - \frac{1}{2} + 2 \right]$$

$$= \left[\frac{16-12-24}{6} \right] - \left[\frac{-2-3+12}{6} \right]$$

$$= \left[\frac{16-36}{6} \right] - \left[\frac{-5+12}{6} \right]$$

$$= -\frac{20}{6} - \frac{7}{6}$$

$$= -\frac{20+7}{6}$$

$$= -\frac{27}{6} \text{ units.}$$

$$\therefore A = \frac{27}{6} \text{ units.}$$

* Find the area of the region bounded by the curve $x^2-7x+10$ and x-axis.

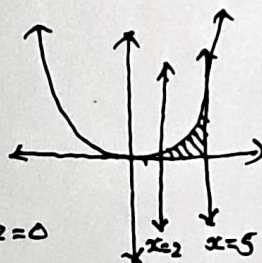
Here $y=x^2-7x+10$

$$\Rightarrow x^2-7x+10=0$$

$$\Rightarrow (x-5)(x-2)=0$$

$$\Rightarrow x-5=0 \text{ or } x-2=0$$

$$\Rightarrow x=5 \text{ or } x=2.$$



$$A = \int_a^b y \, dx$$

$$= \int_2^5 (x^2 - 7x + 10) \, dx$$

$$= \left[\frac{x^3}{3} - 7\frac{x^2}{2} + 10x \right]_2^5$$

$$\begin{aligned}
 &= \left[\frac{(5)^3}{3} - 7 \frac{(5)^2}{2} + 10(5) \right] - \left[\frac{(2)^3}{3} - 7 \frac{(2)^2}{2} + 10(2) \right] \\
 &= \left[\frac{125}{3} - \frac{175}{2} + 50 \right] - \left[\frac{8}{3} - \frac{28}{2} + 20 \right] \\
 &= \left[\frac{250 - 525 + 300}{6} \right] - \left[\frac{16 - 84 + 120}{6} \right] \\
 &= \frac{25}{6} - \frac{52}{6} \\
 &= \frac{25 - 52}{6} \\
 &= -\frac{27}{6}
 \end{aligned}$$

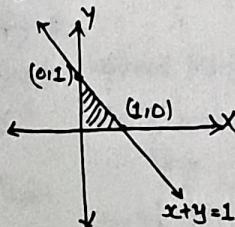
$$A = \frac{27}{6} \text{ units.}$$

* Find the area bounded by the line $x+y=1$ and axes.

$$\Rightarrow x+y=1$$

$$\Rightarrow y=1-x$$

$$x=0, x=1.$$



$$A = \int_a^b y \, dx$$

$$= \int_0^1 (1-x) \, dx$$

$$= \left[x - \frac{x^2}{2} \right]_0^1$$

$$= \left[1 - \frac{(1)^2}{2} \right] - [0 - 0]$$

$$= 1 - \frac{1}{2}$$

$$A = \frac{1}{2} \text{ units.}$$